



# California Regional Water Quality Control Board North Coast Region

Geoffrey M. Hales, Chairman



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Arnold  
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**ORDER NO. R1-2010-0020**  
**NPDES NO. CA0022870**  
**WDID NO. 1B83129OMEN**

## **WASTE DISCHARGE REQUIREMENTS FOR THE MENDOCINO CITY COMMUNITY SERVICES DISTRICT WASTEWATER TREATMENT FACILITY**

The following Discharger is subject to waste discharge requirements as set forth in this Order:

**Table 1. Discharger Information**

|                                                                                                                                                    |                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Discharger</b>                                                                                                                                  | Mendocino City Community Services District (CSD)        |
| <b>Name of Facility</b>                                                                                                                            | Mendocino City CSD Wastewater Treatment Facility (WWTF) |
| <b>Facility Address</b>                                                                                                                            | 10500 Kelly Street                                      |
|                                                                                                                                                    | Mendocino, CA 95460                                     |
|                                                                                                                                                    | Mendocino County                                        |
| The U.S. Environmental Protection Agency (USEPA) and the Regional Water Quality Control Board have classified this discharge as a minor discharge. |                                                         |

Discharges by the Mendocino City CSD from the discharge points identified below are subject to waste discharge requirements as set forth in this Order.

**Table 2. Discharge Location**

| <b>Discharge Point</b> | <b>Effluent Description</b>           | <b>Discharge Point Latitude</b> | <b>Discharge Point Longitude</b> | <b>Receiving Water</b> |
|------------------------|---------------------------------------|---------------------------------|----------------------------------|------------------------|
| 001                    | Tertiary-treated Municipal Wastewater | 39° 18' 21" N                   | 123° 48' 30" W                   | Pacific Ocean          |
| 002                    | Recycled Water                        | ---                             | ---                              | Recycled Water System  |

**Table 3. Administrative Information**

|                                                                                                                                                                                                   |                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| This Order was adopted by the Regional Water Quality Control Board on:                                                                                                                            | April 29, 2010                              |
| This Order shall become effective on:                                                                                                                                                             | July 1, 2010                                |
| This Order shall expire on:                                                                                                                                                                       | July 1, 2015                                |
| The Discharger shall file a Report of Waste Discharge in accordance with title 23, California Code of Regulations, as application for issuance of new waste discharge requirements no later than: | 180 days prior to the Order expiration date |

IT IS HEREBY ORDERED, that Order No. R1-2004-0055 is rescinded upon the effective date of this Order except for enforcement purposes, and, in order to meet the provisions contained in division 7 of the California Water Code (commencing with section 13000) and regulations adopted thereunder, and the provisions of the federal Clean Water Act (CWA) and regulations and guidelines adopted thereunder, the Discharger shall comply with the requirements in this Order.

I, Catherine Kuhlman, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, North Coast Region, on April 29, 2010.

  
for Catherine Kuhlman, Executive Officer

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## I. Facility Information

The following Discharger is subject to waste discharge requirements as set forth in this Order.

**Table 4. Facility Information**

|                                           |                                                                            |
|-------------------------------------------|----------------------------------------------------------------------------|
| <b>Discharger</b>                         | Mendocino City CSD                                                         |
| <b>Name of Facility</b>                   | Mendocino City CSD WWTF                                                    |
| <b>Facility Address</b>                   | 10500 Kelly Street                                                         |
|                                           | Mendocino, CA 95460                                                        |
|                                           | Mendocino County                                                           |
| <b>Facility Contact, Title, Phone No.</b> | Michael Kelley, District Superintendent, (707) 937-5790                    |
| <b>Mailing Address</b>                    | P.O. Box 1029, Mendocino CA 95460                                          |
| <b>Type of Facility</b>                   | Publicly Owned Treatment Works                                             |
| <b>Facility Design Flow</b>               | 0.3 million gallons per day (mgd) (average dry weather treatment capacity) |

## II. Findings

The California Regional Water Quality Control Board, North Coast Region (hereinafter the Regional Water Board), finds:

- A. Background.** The Mendocino City CSD (hereinafter the Discharger) is currently discharging pursuant to Order No. R1-2004-0055 and National Pollutant Discharge Elimination System (NPDES) Permit No. CA0022870. The Discharger submitted a Report of Waste Discharge (ROWD), dated September 8, 2008, and applied for an NPDES permit renewal to discharge tertiary treated wastewater from the Mendocino City CSD WWTF. The application was deemed complete on May 18, 2009.

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and State laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

- B. Facility Description.** The Discharger owns a wastewater collection, treatment, and disposal facility that serves a population of approximately 4,000 from the community of Mendocino, Russian Gulch State Park, and Headlands State Park. The treatment facility has an average dry weather design treatment capacity of 0.3 mgd and treats domestic and commercial wastewater.

Treatment at the Mendocino City CSD WWTF consists of comminution, extended aeration activated sludge treatment, secondary clarification, tertiary dual-media filtration, chlorine disinfection and dechlorination. Biosolids are aerobically digested and dried with a belt filter press and dryer prior to placement in a permitted landfill. Condensate from the sludge dryer is returned to the headworks. Onsite storm water runoff is also diverted to the plant headworks for treatment. Influent flows exceeding the design capacity can be routed to a 300,000 gallon overflow pond for storage until flows diminish and the excess flow can be returned to the headworks for treatment.

Treated effluent flows under gravity forces to a flow equalization pond, which provides for a continuous rate of discharge. Effluent from the equalization pond is controlled at flow control structure A by sliding stop gates, from which flow continues to flow control structure B, which has two pressure control valves to prevent backflow conditions in the outfall. From flow control structure B, effluent is discharged to the Pacific Ocean at Discharge Point 001 via a 996 foot outfall designed to achieve a minimal dilution of 100:1.

During the dry weather season (generally May through October) and other periods as needed, tertiary treated effluent is discharged in batches to the water recycling system, which consists of a 55,000 gallon storage tank, accompanying appurtenances, and a pop-up sprinkler system located on the Mendocino High School athletic fields. The recycled water system is regulated under Waste Reclamation Requirements Order No. 97-66, which contains applicable state and local requirements regarding the production and use of reclaimed wastewater.

Attachment B provides a map of the area around the facility. Attachment C provides a flow schematic of the facility.

- C. Legal Authorities.** This Order is issued pursuant to section 402 of the federal CWA and implementing regulations adopted by the USEPA and chapter 5.5, division 7 of the California Water Code (commencing with section 13370). It shall serve as an NPDES permit for point source discharges from this facility to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to article 4, chapter 4, division 7 of the Water Code (commencing with section 13260).
- D. Background and Rationale for Requirements.** The Regional Water Board developed the requirements in this Order based on information submitted as part of the application, through monitoring and reporting programs, and other available information. The Fact Sheet (Attachment F), which contains background information and rationale for Order requirements, is hereby incorporated into this Order and constitutes part of the Findings for this Order. Attachments A through E are also incorporated into this Order.
- E. California Environmental Quality Act (CEQA).** Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of CEQA, Public Resources Code sections 21100-21177.
- F. Technology-Based Effluent Limitations.** Section 301(b) of the CWA and implementing USEPA permit regulations at section 122.44, title 40 of the Code of Federal Regulations<sup>1</sup>, require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge authorized by this Order must meet minimum federal technology-based requirements based on Secondary Treatment Standards at Part 133 and/or Best Professional Judgment (BPJ) in accordance with Part 125, section 125.3 and technology-based limits set in Table A of the Ocean Plan. A detailed discussion of the technology-based effluent limitations development is included in the Fact Sheet (Attachment F).
- G. Water Quality-Based Effluent Limitations.** Section 301(b) of the CWA and section 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards.

Section 122.44(d)(1)(i) mandates that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, water quality-based effluent limitations (WQBELs) must be established using: (1) USEPA criteria guidance

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<sup>1</sup> All further statutory references are to title 40 of the Code of Federal Regulations unless otherwise indicated.

under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in section 122.44(d)(1)(vi).

**H. Water Quality Control Plans.** The Regional Water Board adopted a *Water Quality Control Plan for the North Coast Region* (hereinafter the Basin Plan) that designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. In addition, the Basin Plan implements State Water Resources Control Board (State Water Board) Resolution No. 88-63, which establishes State policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply. With total dissolved solid concentrations much greater than 3,000 mg/L, ocean waters meet an exception to Resolution 88-63; and therefore, the "municipal or domestic supply" (MUN) designation is not applicable to the ocean receiving water for the discharge at Discharge Point 001. Beneficial uses established by the Basin Plan for ocean waters are described in Table 5, below.

**Table 5. Basin Plan Beneficial Uses**

| Discharge Point | Receiving Water | Beneficial Uses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 001             | Pacific Ocean   | <p><b>Existing:</b></p> <ul style="list-style-type: none"> <li>• Navigation (NAV)</li> <li>• Water Contact Recreation (REC1)</li> <li>• Non-Contact Water Recreation (REC2)</li> <li>• Commercial and Sport Fishing (COMM)</li> <li>• Marine Habitat (MAR)</li> <li>• Wildlife Habitat (WILD)</li> <li>• Preservation of Rare, Threatened, or Endangered Species (RARE)</li> <li>• Migration of Aquatic Organisms (MIGR)</li> <li>• Spawning, Reproduction, and/or Early Development (SPWN)</li> <li>• Shellfish Harvesting (SHELL)</li> <li>• Aquaculture (AQUA)</li> </ul> <p><b>Potential:</b></p> <ul style="list-style-type: none"> <li>• Industrial Service Supply (IND)</li> <li>• Industrial Process Supply (PRO)</li> <li>• Preservation of Areas of Special Biological Significance (ASBS)</li> </ul> |

Requirements of this Order implement the Basin Plan.

The State Water Board adopted the *Water Quality Control Plan for Control of Temperature in the Coastal and Interstate Water and Enclosed Bays and Estuaries of California* (Thermal Plan) on May 18, 1972, and amended this plan on September 18, 1975. This plan contains temperature objectives for coastal waters. Requirements of this Order implement the Thermal Plan.

- I. California Ocean Plan.** The State Water Board adopted the *Water Quality Control Plan for Ocean Waters of California*, California Ocean Plan (Ocean Plan) in 1972 and amended it in 1978, 1983, 1988, 1990, 1997, 2000, and 2005. The State Water Board adopted the latest amendments on April 21, 2005 and it became effective on February 14, 2006. The Ocean Plan is applicable, in its entirety, to point source discharges to the Pacific Ocean. The Ocean Plan identifies beneficial uses of ocean waters of the State to be protected as summarized below.

**Table 6. Ocean Plan Beneficial Uses**

| Discharge Point | Receiving Water | Beneficial Uses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 001             | Pacific Ocean   | <ul style="list-style-type: none"> <li>• Industrial Water Supply;</li> <li>• Water Contact and Non-Contact Recreation, Including Aesthetic Enjoyment;</li> <li>• Navigation;</li> <li>• Commercial and Sport Fishing;</li> <li>• Mariculture;</li> <li>• Preservation and Enhancement of Designated Areas of Special Biological Significance (ASBS);</li> <li>• Rare and Endangered Species;</li> <li>• Marine Habitat;</li> <li>• Fish Migration</li> <li>• Fish Spawning; and</li> <li>• Shellfish Harvesting</li> </ul> |

In order to protect beneficial uses, the Ocean Plan establishes water quality objectives and a program of implementation. Requirements of this Order implement the Ocean Plan.

- J. Alaska Rule.** On March 30, 2000, USEPA revised its regulation that specifies when new and revised state and tribal water quality standards (WQS) become effective for CWA purposes. [40 C.F.R. § 131.21; 65 Fed. Reg. 24641 (April 27, 2000)] Under the revised regulation (also known as the Alaska Rule), new and revised standards submitted to USEPA after May 30, 2000, must be approved by USEPA before being used for CWA purposes. The final rule also provides that standards already in effect and submitted to USEPA by May 30, 2000 may be used for CWA purposes, whether or not approved by USEPA.

**K. Stringency of Requirements for Individual Pollutants.** This Order contains both technology-based and water quality-based effluent limitations for individual pollutants. The technology-based effluent limitations consist of restrictions on pH, biochemical oxygen demand (BOD<sub>5</sub>), total suspended solids (TSS), settleable solids, turbidity, and oil and grease at Discharge Point 001. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements. In addition, this Order contains effluent limitations more stringent than the minimum, federal technology-based requirements that are necessary to meet water quality standards. These limitations are not more stringent than required by the CWA.

Water quality-based effluent limitations for total residual chlorine, chloroform, dichlorobromomethane, TCDD Equivalents and total coliform bacteria have been scientifically derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. The scientific procedures for calculating the individual water quality-based effluent limitations are based on the Ocean Plan, which was approved by USEPA on February 14, 2006. All beneficial uses and water quality objectives contained in the Basin Plan were approved under State law and submitted to and approved by USEPA prior to May 30, 2000. Any water quality objectives and beneficial uses submitted to USEPA prior to May 30, 2000, but not approved by USEPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to section 131.21(c)(1). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the requirements of the CWA.

**L. Antidegradation Policy.** Section 131.12 requires that state water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution No. 68-16. Resolution No. 68-16 incorporates the federal antidegradation policy where the federal policy applies under federal law. Resolution No. 68-16 requires that existing quality of waters be maintained unless degradation is justified based on specific findings. The Regional Water Board's Basin Plan implements, and incorporates by reference, both the State and federal antidegradation policies. As discussed in detail in the Fact Sheet the permitted discharge is consistent with the antidegradation provisions of section 131.12 and State Water Board Resolution No. 68-16.

**M. Anti-Backsliding Requirements.** Sections 402(o)(2) and 303(d)(4) of the CWA and federal regulations section 122.44(l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require effluent limitations in a reissued permit to be as stringent as those in the previous permit, with some exceptions where limitations may be relaxed. All effluent limitations in this Order are at least as stringent that those in the previous Order.

**N. Endangered Species Act.** This Order does not authorize any act that results in the taking of a threatened or endangered species or any act that is now prohibited, or becomes prohibited in the future, under either the California Endangered Species Act

(Fish and Game Code sections 2050 to 2097) or the federal Endangered Species Act (16 U.S.C.A. sections 1531 to 1544). This Order requires compliance with effluent limits, receiving water limits, and other requirements to protect the beneficial uses of waters of the State. The Discharger is responsible for meeting all requirements of the applicable Endangered Species Act.

- O. Monitoring and Reporting.** Section 122.48 requires that all NPDES permits specify requirements for recording and reporting monitoring results. Water Code sections 13267 and 13383 authorize the Regional Water Board to require technical and monitoring reports. The Monitoring and Reporting Program (MRP) establishes monitoring and reporting requirements to implement federal and State requirements. This Monitoring and Reporting Program is provided in Attachment E.
- P. Standard and Special Provisions.** Standard Provisions, which apply to all NPDES permits in accordance with section 122.41, and additional conditions applicable to specified categories of permits in accordance with section 122.42, are provided in Attachment D. The Discharger must comply with all standard provisions and with those additional conditions that are applicable under section 122.42. The Regional Water Board has also included in this Order special provisions applicable to the Discharger. A rationale for the special provisions contained in this Order is provided in the attached Fact Sheet.
- Q. Provisions and Requirements Implementing State Law.** The provisions/requirements in subsection V.B of this Order are included to implement State law only. These provisions/requirements are not required or authorized under the federal CWA; consequently, violations of these provisions/requirements are not subject to the enforcement remedies that are available for NPDES violations.
- R. Notification of Interested Parties.** The Regional Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe Waste Discharge Requirements for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Details of notification are provided in the Fact Sheet of this Order.
- S. Consideration of Public Comment.** The Regional Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Public Hearing process are provided in the Fact Sheet of this Order.

### **III. Discharge Prohibitions**

- A. The discharge of any waste not disclosed by the Discharger or not within the reasonable contemplation of the Regional Water Board is prohibited.
- B. Creation of pollution, contamination, or nuisance, as defined by Section 13050 of the California Water Code, is prohibited.
- C. The discharge of sludge or digester supernatant is prohibited, except as authorized under section VI.C.5.c of this Order (Sludge Disposal and Handling Requirements).
- D. The discharge of untreated or partially treated waste (receiving a lower level of treatment than described in Findings II.B of the Order) from anywhere within the collection, treatment, or disposal systems is prohibited, except as provided for in Prohibition III. E and Attachment D, Standard Provision G (Bypass).
- E. Any sanitary sewer overflow (SSO) that results in a discharge of untreated or partially treated wastewater to (a) waters of the State, (b) groundwater, or (c) land, that creates pollution, contamination, or nuisance, as defined in Water Code section 13050 (m) is prohibited.
- F. The discharge of more than 0.3 mgd as an average dry weather flow, as determined from the lowest average daily flow measured over 30 consecutive days, is prohibited.
- G. The discharge of recycled, filtered wastewater to a use area other than those designated for that purpose is prohibited.
- H. The discharge of any radiological, chemical, or biological warfare agent or high-level radioactive waste into the ocean is prohibited.
- I. The discharge of sludge directly into the ocean or into a waste stream that discharges to the ocean is prohibited.

#### IV. Effluent Limitations and Discharge Specifications

##### A. Final Effluent Limitations – Discharge Point 001

###### 1. Final Effluent Limitations – Discharge Point 001

- a. The Discharger shall maintain compliance with the following final effluent limitations at Discharge Point 001, with compliance measured at Monitoring Location EFF-001, as described in the attached MRP (Attachment E).

**Table 7. Effluent Limitations for Discharge Point 001**

| Parameter               | Units                | Effluent Limitations                   |                |               |                       |                  |
|-------------------------|----------------------|----------------------------------------|----------------|---------------|-----------------------|------------------|
|                         |                      | Average Monthly                        | Average Weekly | Maximum Daily | Instantaneous Maximum | Six-Month Median |
| BOD <sub>5</sub>        | mg/L                 | 30                                     | 45             | ---           | ---                   | ---              |
|                         | lbs/day <sup>2</sup> | 75                                     | 113            | ---           | ---                   | ---              |
| TSS                     | mg/L                 | 30                                     | 45             | ---           | ---                   | ---              |
|                         | lbs/day <sup>2</sup> | 75                                     | 113            | ---           | ---                   | ---              |
| pH                      | s.u.                 | Not less than 6.0 nor greater than 9.0 |                |               |                       |                  |
| Settleable Solids       | mL/L-hr              | 1.0                                    | 1.5            | 3.0           | ---                   | ---              |
| Oil and Grease          | mg/L                 | 25                                     | 40             | 75            | ---                   | ---              |
| Turbidity               | NTU                  | 75                                     | 100            | 225           | ---                   | ---              |
| Total Chlorine Residual | mg/L                 | ---                                    | ---            | 0.81          | 6.06                  | 0.20             |
|                         | lbs/day <sup>2</sup> | ---                                    | ---            | 2.02          | 15.2                  | 0.50             |
| TCDD Equivalents        | µg/L                 | 3.94E-07                               | ---            | ---           | ---                   | ---              |
|                         | lbs/day <sup>2</sup> | 9.86E-10                               | ---            | ---           | ---                   | ---              |
| Chloroform              | µg/L                 | 13,130                                 | ---            | ---           | ---                   | ---              |
|                         | lbs/day <sup>2</sup> | 32.9                                   | ---            | ---           | ---                   | ---              |
| Dichlorobromo-methane   | µg/L                 | 626                                    | ---            | ---           | ---                   | ---              |
|                         | lbs/day <sup>2</sup> | 1.57                                   | ---            | ---           | ---                   | ---              |

- b. **Percent Removal:** The average monthly percent removal of BOD<sub>5</sub> and TSS shall not be less than 85 percent, determined by comparing the average monthly influent concentration to the average monthly effluent concentration, as measured at Monitoring Locations INF-001 and EFF-001.
- c. **Total Coliform Bacteria:** Disinfected effluent discharged at Discharge Point 001 shall not contain total coliform bacteria in excess of the following limitations:

<sup>2</sup> Mass-based effluent limitations are based on the average dry weather design flow of 0.3 mgd.

- (1) The median value of total coliform bacteria shall not exceed a Most Probable Number (MPN) of 70 per 100 milliliters (mL) in a calendar month, and
- (2) In not more than 10 percent of samples shall total coliform bacteria exceed an MPN of 230 per 100 mL, using bacteriological results from ten consecutive samples.

## B. Land Discharge Specifications

Not Applicable.

## C. Reclamation Specifications – Discharge Point 002

### 1. Final Effluent Limitations – Discharge Point 002

- a. The Discharger shall maintain compliance with the following effluent limitations when discharging to the water recycling system, with compliance measured at monitoring locations described in the attached MRP:

**Table 8. Effluent Limitations for Discharge Point 002**

| Parameter        | Units    | Effluent Limitation |               |
|------------------|----------|---------------------|---------------|
|                  |          | Average Monthly     | Maximum Daily |
| BOD <sub>5</sub> | mg/L     | 10                  | 20            |
| TSS              | mg/L     | 10                  | 20            |
| Nitrate (as N)   | mg/L     | 10                  | ---           |
| pH               | pH Units | 6.0 – 9.0           |               |

- b. **Bacteria:** Disinfected effluent discharged to the water recycling system shall not contain coliform bacteria in excess of the following concentrations, with compliance measured at Monitoring Location EFF-002, as described in the attached MRP:
  - (1) The median concentration shall not exceed a Most Probable Number (MPN) of 2.2 per 100 milliliters, using the bacteriological results of the last seven days for which analyses have been completed,
  - (2) The number of coliform bacteria shall not exceed an MPN of 23 per 100 milliliters in more than one sample in any 30-day period, and
  - (3) No single sample shall exceed an MPN of 240 total coliform bacteria per 100 milliliters.
- c. **Turbidity.** The effluent from the filtration system shall at all times be filtered such that the filtered effluent does not exceed any of the following specifications prior to discharge to the disinfection unit, with compliance measured at Monitoring Location EFF-001, as described in the attached MRP:

- (1) An average of 2 Nephelometric Turbidity Units (NTU) during a 24-hour period;
  - (2) 5 NTU more than 5 percent of the time during a 24-hour period; and
  - (3) 10 NTU at any time.
- c. **Filtration Rate.** The rate of filtration through the effluent filter shall not exceed five (5) gallons per minute per square foot of surface area or other filtration rates authorized in writing by the Executive Officer and under conditions recommended by the California Department of Public Health, with compliance as described in the attached MRP.
- d. **Disinfection Process Requirements.** Treated effluent shall be disinfected in a manner that ensures effective pathogen reduction as described in the following specifications, with compliance measured at monitoring locations described in the attached MRP:
- (1) Before recycled water is used to irrigate the athletic fields, a CT value<sup>3</sup> of not less than 450 milligram-minutes per liter shall be achieved by maintaining a chlorine residual in the recycled water storage tank for a predetermined contact time. The minimum allowable contact time, between the time pumping from the WWTF is stopped and the time irrigation begins, is 90 minutes.
  - (2) Recycled water not meeting the CT criteria shall not be discharged for irrigation at the athletic fields.

## V. Receiving Water Limitations

### A. Surface Water Limitations

The following receiving water conditions are based on water quality objectives established by the Ocean Plan and are a required part of this Order. The discharge of waste shall not cause the following water quality objectives to be violated upon completion of final dilution.

#### *Ocean Plan*

1. Bacterial Characteristics
  - a. Body Contact Standards

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<sup>3</sup> The CT value is the product of total chlorine residual and modal contact time measured at the same period. The modal contact time is the amount of time that elapsed between the time that a tracer, such as salt or dye, is injected into the influent at the entrance of the chlorination chamber and the time that the highest concentration of the tracer is observed in the effluent from the chamber.

Within a zone bounded by the shoreline and a distance of 1,000 feet from the shoreline or the 30-foot depth contour, whichever is further from the shoreline, and in areas outside this zone used for water contact sports, as determined by the Regional Water Board (i.e., waters designated as REC-1), but including all kelp beds, the following bacteriological objectives shall be maintained throughout the water column:

30-Day Geometric Mean – The following standards are based on the geometric mean of the five most recent samples from each receiving water monitoring location:

- i. Total coliform density shall not exceed 1,000 per 100 mL
- ii. Fecal coliform density shall not exceed 200 per 100 mL; and
- iii. Enterococcus density shall not exceed 35 per 100 mL.

Single Sample maximum:

- i. Total coliform density shall not exceed 10,000 per 100 mL;
- ii. Fecal coliform density shall not exceed 400 per 100 mL;
- iii. Enterococcus density shall not exceed 104 per 100 mL; and
- iv. Total coliform density shall not exceed 1,000 per 100 mL when the fecal coliform to total coliform ratio exceeds 0.1

b. Shellfish Harvesting Standards

At all areas where shellfish may be harvested for human consumption, as determined by the Regional Water Board, the following bacteriological objectives shall be maintained throughout the water column:

- i. The median total coliform density shall not exceed 70 organisms per 100 mL, and in not more than 10 percent of samples shall coliform density exceed 230 organisms per 100 mL.

2. Physical Characteristics

- a. Floating particulates and grease and oil shall not be visible.
- b. The discharge of waste shall not cause aesthetically undesirable discoloration of the ocean surface.
- c. Natural light shall not be significantly reduced at any point outside the initial dilution zone as the result of the discharge of waste.
- d. The rate of deposition of inert solids and the characteristics of inert solids in ocean sediments shall not be changed such that benthic communities are degraded.

### 3. Chemical Characteristics

- a. The dissolved oxygen concentration shall not at any time be depressed more than 10 percent from that which occurs naturally as a result of the discharge of oxygen demanding waste material.
- b. The pH shall not be changed at any time more than 0.2 units from that which occurs naturally.
- c. The dissolved sulfide concentration of waters in and near sediments shall not be significantly increased above that present under natural conditions.
- d. The concentration of substances set forth in Chapter IV, Table B of the Ocean Plan in marine sediments shall not be increased to levels that would degrade indigenous biota.
- e. The concentration of organic materials in marine sediments shall not be increased to levels that would degrade marine life.
- f. Nutrient levels shall not cause objectionable aquatic growths or degrade indigenous biota.
- g. Discharges shall not cause exceedances of water quality objectives for ocean waters of the State established in Table B of the Ocean Plan.
- h. Discharge of radioactive waste shall not degrade marine life.

### 4. Biological Characteristics

- a. Marine communities, including vertebrate, invertebrate and plant species, shall not be degraded.
- b. The natural taste, odor, and color of fish, shellfish, or other marine resources used for human consumption shall not be altered.
- c. The concentration of organic materials in fish, shellfish, or other marine resources used for human consumption shall not bioaccumulate to levels that are harmful to human health.

### 5. General Standards

- a. The discharge shall not cause a violation of any applicable water quality standard for the receiving waters adopted by the Regional Water Board or the State Water Board as required by the Clean Water Act and regulations adopted thereunder.
- b. The discharge shall be essentially free of:
  - i. Material that is floatable or will become floatable upon discharge.
  - ii. Settleable material or substances that may form sediments that will degrade benthic communities or other aquatic life.
  - iii. Substances that will accumulate to toxic levels in marine waters, sediments or biota.

- iv. Substances that significantly decrease natural light to benthic communities and other marine life.
- v. Material that results in aesthetically undesirable discoloration of the ocean surface.
- c. Waste effluent shall be discharged in a manner that provides sufficient initial dilution to minimize the concentrations of substances not removed in the treatment.
- d. Location of waste discharges must be determined after a detailed assessment of the oceanographic characteristics and current patterns to assure that:
  - i. Pathogenic organisms and viruses are not present in areas where shellfish are harvested for human consumption or in areas used for swimming or other body contact sports.
  - ii. Natural water quality conditions are not altered in areas designated as being of special biological significance.
  - iii. Maximum protection is provided to the marine environment.
  - iv. The discharge does not adversely affect recreational beneficial uses such as surfing and beach walking.

## **B. Groundwater Limitations**

- 1. The collection, storage, and use of wastewater shall not cause or contribute to a statistically significant degradation of groundwater quality.
- 2. The collection, storage, and use of wastewater shall not cause groundwater to contain taste or odor producing substances in concentrations that cause nuisance or adversely affect beneficial uses.

## **VI. Provisions**

### **A. Standard Provisions**

- 1. **Federal Standard Provisions.** The Discharger shall comply with all Standard Provisions included in Attachment D of this Order.
- 2. **Regional Water Board Standard Provisions.** The Discharger shall comply with the following Regional Water Board standard provisions.
  - a. Failure to comply with provisions or requirements of this Order, or violation of other applicable laws or regulations governing discharges from this facility, may subject the Discharger to administrative or civil liabilities, criminal penalties, and/or other enforcement remedies to ensure compliance. Additionally, certain violations may subject the Discharger to civil or criminal enforcement from appropriate local, state, or federal law enforcement entities.

- b. In the event the Discharger does not comply or will be unable to comply for any reason, with any prohibition, interim or final effluent limitation, land discharge specification, reclamation specification, receiving water limitation, or provision of this Order that may result in a significant threat to human health or the environment, such as inundation of treatment components, breach of pond containment, sanitary sewer overflow, irrigation runoff, etc., that results in a discharge to a drainage channel or a surface water, the Discharger shall as soon as possible, but no later than two (2) hours after becoming aware of the discharge, orally<sup>4</sup> notify the State Office of Emergency Services, the local health officer or directors of environmental health with jurisdiction over the affected water bodies, and the Regional Water Board.
- c. As soon as possible, but no later than twenty-four (24) hours after becoming aware of an unauthorized discharge to a drainage channel or a surface water, the Discharger shall submit to the Regional Water Board a written certification that the State Office of Emergency Services and the local health officer or directors of environmental health with jurisdiction over the affected water body have been notified of the discharge. Written documentation of the circumstances of the spill event shall be submitted to the Regional Water Board within five (5) days, unless the Regional Water Board waives the confirmation. The written notification shall state the nature, time, duration, and cause of noncompliance, and shall describe the measures being taken to remedy the current noncompliance and to prevent recurrence, including, where applicable, a schedule of implementation. Other types of noncompliance require written notification, as described above, at the time of the normal monitoring report.

## **B. Monitoring and Reporting Program (MRP) Requirements**

The Discharger shall comply with the MRP, and future revisions thereto, in Attachment E of this Order.

## **C. Special Provisions**

### **1. Reopener Provisions**

- a. **Standard Revisions.** If applicable water quality standards are promulgated or approved pursuant to Section 303 of the CWA, or amendments thereto, the Regional Water Board may reopen this Order and make modifications in accordance with such revised standards.

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<sup>3</sup> Oral reporting means direct contact with a Regional Water Board staff person. The oral report may be given in person or by telephone. After business hours, oral contact must be made by calling the State Office of Emergency Services at (800) 852-7550. After normal business hours, spill reporting to OES will satisfy the 2 hour notification requirement for the Regional Water Board.

- b. Reasonable Potential.** This Order may be reopened for modification to include an effluent limitation if monitoring establishes that the discharge causes, or has the reasonable potential to cause or contribute to, an excursion above a water quality criterion or objective applicable to the receiving water.
- c. Whole Effluent Toxicity.** As a result of a Toxicity Reduction Evaluation (TRE), this Order may be reopened to include a limitation for a specific toxicant identified in the TRE.

## **2. Special Studies, Technical Reports and Additional Monitoring Requirements**

### **a. Toxicity Reduction Requirements**

- (1) Whole Effluent Toxicity.** The Monitoring and Reporting Program (MRP) of this Order requires routine monitoring for chronic toxicity at Discharge Point 001 to determine compliance with the Ocean Plan's water quality objective for chronic toxicity. As established by the MRP, if chronic toxicity is measured in the effluent above 101 TUC, the Discharger shall conduct accelerated monitoring as specified in section V. of the MRP.

Results of accelerated toxicity monitoring will indicate a need to conduct a Toxicity Reduction Evaluation (TRE), if toxicity persists; or it will indicate that a return to routine toxicity monitoring is justified because persistent toxicity has not been identified by accelerated monitoring. TREs shall be conducted in accordance with the TRE Workplan prepared by the Discharger pursuant to Section VI.C.2.a(2) of this Order, below.

- (2) Toxicity Reduction Evaluations (TRE) Workplan.** The Discharger shall maintain a TRE Workplan. This plan shall be reviewed and updated as necessary in order to remain current and applicable to the discharge and discharge facilities. The Workplan shall describe the steps the Discharger intends to follow if toxicity is detected above 101 TUC, and should include at least the following items:
  - (a) A description of the investigation and evaluation techniques that would be used to identify potential causes and sources of toxicity, effluent variability, and treatment system efficiency.
  - (b) A description of the facility's methods of maximizing in house treatment efficiency and good housekeeping practices.
  - (c) If a toxicity identification evaluation (TIE) is necessary, an indication of the person who would conduct the TIEs (i.e., an in house expert or an outside contractor).

- (3) Toxicity Reduction Evaluations (TRE).** The TRE shall be conducted in accordance with the following:

- (a) The TRE shall be initiated within 30 days of the date of completion of the accelerated monitoring test, required by Section V of the MRP, observed to exceed either the acute or chronic toxicity parameter.
- (b) The TRE shall be conducted in accordance with the Discharger's Workplan.
- (c) The TRE shall be in accordance with current technical guidance and reference material including, at a minimum, the USEPA manual EPA/833B 99/002.
- (d) The TRE may end at any stage if, through monitoring results, it is determined that there is no longer consistent toxicity.
- (e) The Discharger may initiate a TIE as part of the TRE process to identify the cause(s) of toxicity. As guidance, the Discharger shall use the USEPA acute and chronic manuals, EPA/600/6-91/005F (Phase I), EPA/600/R-92/080 (Phase II), and EPA-600/R-92/081 (Phase III).
- (f) As toxic substances are identified or characterized, the Discharger shall continue the TRE by determining the source(s) and evaluating alternative strategies for reducing or eliminating the substances from the discharge. All reasonable steps shall be taken to reduce toxicity to levels consistent with acute and chronic toxicity parameters.
- (g) Many recommended TRE elements accompany required efforts of source control, pollution prevention, and storm water control programs. TRE efforts should be coordinated with such efforts. To prevent duplication of efforts, evidence of complying with requirements of recommendations of such programs may be acceptable to comply with requirements of the TRE.
- (h) The Regional Water Board recognizes that chronic toxicity may be episodic and identification of a reduction of sources of chronic toxicity may not be successful in all cases. Consideration of enforcement action by the Regional Water Board will be based in part on the Discharger's actions and efforts to identify and control or reduce sources of consistent toxicity.

### **3. Best Management Practices and Pollution Prevention**

#### **a. Pollutant Minimization Program (PMP)**

The Discharger shall, as required by the Executive Officer, develop and conduct a PMP as further described below when there is evidence (e.g., sample results reported as detected, not quantified (DNQ) when the effluent limitation is less than the minimum detection limit (MDL), sample results from analytical methods more sensitive than those methods required by this Order, presence of whole effluent toxicity, health advisories for fish consumption, results of benthic or

aquatic organism tissue sampling) that a priority pollutant is present in the effluent above an effluent limitation and either:

- (1) The concentration of the pollutant is reported as DNQ and the effluent limitation is less than the reported Minimum Level (ML); or
- (2) The concentration of the pollutant is reported as Not Detected (ND) and the effluent limitation is less than the MDL, using definitions described in Attachment A and reporting protocols described in MRP section X.B.4.

The PMP shall include, but not be limited to, the following actions and submittals acceptable to the Regional Water Board:

- (1) An annual review and semi-annual monitoring of potential sources of the reportable priority pollutant(s), which may include fish tissue monitoring and other bio-uptake sampling;
- (2) Quarterly monitoring for the reportable priority pollutant(s) in the influent to the wastewater treatment system;
- (3) Submittal of a control strategy designed to proceed toward the goal of maintaining concentrations of the reportable priority pollutant(s) in the effluent at or below the effluent limitation;
- (4) Implementation of appropriate cost-effective control measures for the reportable priority pollutant(s), consistent with the control strategy; and
- (5) An annual status report that shall be sent to the Regional Water Board including:
  - (a) All PMP monitoring results for the previous year;
  - (b) A list of potential sources of the reportable priority pollutant(s);
  - (c) A summary of all actions undertaken pursuant to the control strategy; and
  - (d) A description of actions to be taken in the following year.

#### **4. Construction, Operation and Maintenance Specifications**

##### **a. Operation and Maintenance (O&M) Manual.**

The Discharger shall maintain an updated O&M Manual for the Facility. The Discharger shall update the O&M Manual, as necessary, to conform with changes in operation and maintenance of the Facility. The O&M Manual shall be readily available to operating personnel onsite. The O&M Manual shall include the following:

- (1) Description of the treatment plant table of organization showing the number of employees, duties and qualifications and plant attendance schedules (daily, weekends and holidays, part-time, etc). The description should include documentation that the personnel are knowledgeable and qualified to operate the treatment facility so as to achieve the required level of treatment at all times.
- (2) Detailed description of safe and effective operation and maintenance of treatment processes, process control instrumentation and equipment.
- (3) Description of laboratory and quality assurance procedures.
- (4) Process and equipment inspection and maintenance schedules.
- (5) Description of safeguards to assure that, should there be reduction, loss, or failure of electric power, the Discharger will be able to comply with requirements of this Order.
- (6) Description of preventive (fail-safe) and contingency (response and cleanup) plans for controlling accidental discharges, and for minimizing the effect of such events. These plans shall identify the possible sources (such as loading and storage areas, power outage, waste treatment unit failure, process equipment failure, tank and piping failure) of accidental discharges, untreated or partially treated waste bypass, and polluted drainage.

## **5. Special Provisions for Municipal Facilities (POTWs Only)**

### **a. Wastewater Collection Systems**

#### **(1) Statewide General WDRs for Sanitary Sewer Systems**

On May 2, 2006, the State Water Board adopted State Water Board Order No. 2006-0003-DWQ, Statewide General WDRs for Sanitary Sewer Systems. Order No. 2006-0003-DWQ requires that all public agencies that currently own or operate sanitary sewer systems apply for coverage under the General WDRs. The deadline for dischargers to apply for coverage under State Water Board Order No. 2006-0003-DWQ was November 6, 2006. The Discharger shall maintain coverage under, and shall be subject to the requirements of Order No. 2006-0003-DWQ and any future revisions thereto for operation of its wastewater collection system.

In addition to the coverage obtained under Order No. 2006-0003-DWQ, the Discharger's collection system is part of the treatment system that is subject to this Order. As such, pursuant to federal regulations, the Discharger must properly operate and maintain its collection system [40 CFR 122.41(e)], report any non-compliance [40 CFR 122.41(l)(6) and (7)], and mitigate any discharge from the collection system in violation of this Order [40 CFR 122.41(d)].

(2) Spills and Sanitary Sewer Overflows

- (a) The Discharger shall take all feasible steps to stop spills and sanitary sewer overflows (SSOs) as soon as possible. All reasonable steps should be taken to collect spilled material and protect the public from contact with wastes or waste-contaminated soil or surfaces.
- (b) The Discharger shall report orally and in writing to the Regional Water Board staff all SSOs and unauthorized spills of waste. Spill notification and reporting shall be conducted in accordance with the Monitoring and Reporting Program.

**b. Source Control Provisions**

The Discharger shall perform source control functions and provide a summary of source control activities conducted in the previous year in the Annual WWTF Report (due by **March 1<sup>st</sup>** to the Regional Water Board). Source Control functions shall include the following:

- (1) Implement the necessary legal authorities to monitor and enforce source control standards, restrict discharges of toxic materials to the collection system and inspect facilities connected to the system.
- (2) Conduct a waste survey at least **every five years** to identify all dischargers that might discharge pollutants that could pass through or interfere with the operation or performance of the Facility.
- (3) Perform public outreach to educate industrial, commercial, and residential users about the importance of preventing discharges of industrial and toxic wastes to the wastewater treatment plant.
- (4) Perform ongoing inspections and monitoring, as necessary, to ensure adequate source control.
- (5) If waste haulers are allowed to discharge to the Facility, establish a waste hauler permit system, to be reviewed by the Executive Officer, to regulate waste haulers discharging to the collection system or Facility.

The annual report summary shall include:

- (1) A copy of the source control standards;
- (2) A description of the waste hauler permit system, if applicable;
- (3) A summary if the compliance and enforcement activities during the past year; and
- (4) A summary of public participation activities.

**c. Sludge Disposal and Handling Requirements**

- (1) Sludge, as used in this Order, means the solid, semisolid, and liquid residues removed during primary, secondary, or advanced wastewater treatment processes. Solid waste refers to grit and screenings generated during preliminary treatment. Biosolids refers to sludge that has been treated, tested, and demonstrated to be capable of being beneficially and legally used pursuant to federal and State regulations as a soil amendment for agriculture, silviculture, horticulture, and land reclamation activities.
- (2) All collected sludges and other solid waste removed from liquid wastes shall be removed from screens, sumps, ponds, and tanks as needed to ensure optimal plant operation and disposed of in accordance with applicable federal and State regulations.
- (3) The use and disposal of biosolids shall comply with all the requirements in section 503, which are enforceable by the USEPA, not the Regional Water Board. If during the life of this Order, the State accepts primacy for implementation of 40 CFR 503, the Regional Water Board may also initiate enforcement where appropriate.
- (4) Sludge or biosolids that are disposed of in a municipal solid waste landfill or used as daily landfill cover shall meet the applicable requirements of section 258. In the annual self-monitoring report, the Discharger shall report the amount of sludge placed in a landfill and the landfill(s) which received the sludge or biosolids.
- (5) The beneficial use of biosolids by application to land as soil amendment is not covered or authorized by this Order. Class B biosolids that are applied to land as soil amendment by the Discharger within the North Coast Region shall comply with State Water Board Water Quality Order No. 2004-0012-DWQ (General Waste Discharge Requirements for the Discharge of Biosolids to Land as a Soil Amendment in Agricultural, Silvicultural, Horticultural, and Land Reclamation Activities) or other WDRs issued by the Regional Water Board.
- (6) The Discharger shall take all reasonable steps to prevent and minimize any sludge use or disposal in violation of this Order that has a likelihood of adversely affecting human health or the environment.
- (7) Solids and sludge treatment, storage, and disposal or reuse shall not create a nuisance, such as objectionable odors or flies, and shall not result in groundwater contamination.
- (8) Solids and sludge treatment and storage sites shall have facilities adequate to divert surface water runoff from adjacent areas, to protect the boundaries of the site from erosion, and to prevent drainage from the treatment and storage

site. Adequate protection is defined as protection from at least a 100-year storm.

- (9) The discharge of sewage sludge and solids shall not cause waste material to be in a position where it is, or can be, conveyed from the treatment and storage sites and deposited in the waters of the State.

**d. Operator Certification.**

Supervisors and operators of municipal wastewater treatment plants shall possess a certificate of appropriate grade in accordance with Title 23, California Code of Regulations, section 3680. The State Water Board may accept experience in lieu of qualification training. In lieu of a properly certified WWTP operator, the State Water Board may approve use of a water treatment plant operator of appropriate grade certified by the State Department of Public Health where water reclamation is involved.

**e. Adequate Capacity**

If the WWTF will reach capacity within 4 years, the Discharger shall notify the Regional Water Board. A copy of such notification shall be sent to appropriate local elected officials, local permitting agencies, and the press. Factors to be evaluated in assessing reserve capacity shall include, at a minimum, (1) comparison of the wet weather design flow with the highest daily flow, and (2) comparison of the average dry weather design flow with the lowest 30-day flow. The Discharger shall demonstrate that adequate steps are being taken to address the capacity problem. The Discharger shall submit a technical report to the Regional Water Board showing how flow volumes will be prevented from exceeding capacity, or how capacity will be increased, within 120 days after providing notification to the Regional Water Board, or within 120 days after receipt of Regional Water Board notification, that the WWTF will reach capacity within four years. The time for filing the required technical report may be extended by the Regional Water Board. An extension of 30 days may be granted by the Executive Officer, and longer extensions may be granted by the Regional Water Board itself. [California Code of Regulations, title 23, section 2232]

**6. Other Special Provisions**

**a. Storm Water**

For the control of storm water discharged from the site of the wastewater treatment plant, if applicable, the Discharger shall obtain authorization to discharge under and meet the requirements of the State Water Board's Water Quality Order 97-03-DWQ, NPDES General Permit No. CAS000001, Waste Discharge Requirements for Discharges of Storm Water Associated with

Industrial Activities Excluding Construction Activities (or subsequent renewed versions of the General Permit).

## **7. Compliance Schedules**

Not Applicable.

## **VII. Compliance Determination**

Compliance with the effluent limitations contained in Section IV of this Order that are derived from Ocean Plan Table B water quality objectives shall be determined as specified below:

### **A. Compliance with Single-Constituent Effluent Limitations.**

Dischargers are out of compliance with the effluent limitation if the concentration of the pollutant (see Section c, below) in the monitoring sample is greater than the effluent limitation and greater than or equal to the reported ML.

### **B. Compliance with Effluent Limitations expressed as a Sum of Several Constituents**

Dischargers are out of compliance with an effluent limitation which applies to the sum of a group of chemicals (e.g., PCB's) if the sum of the individual pollutant concentrations is greater than the effluent limitation. Individual pollutants of the group will be considered to have a concentration of zero if the constituent is reported as ND or DNQ.

### **C. Multiple Sample Data Reduction**

The concentration of the pollutant in the effluent may be estimated from the result of a single sample analysis or by a measure of central tendency (arithmetic mean, geometric mean, median, etc.) of multiple sample analyses when all sample results are quantifiable (i.e., greater than or equal to the reported Minimum Level). When one or more sample results are reported as ND or DNQ, the central tendency concentration of the pollutant shall be the median (middle) value of the multiple samples. If, in an even number of samples, one or both of the middle values is ND or DNQ, the median will be the lower of the two middle values.

## ATTACHMENT A – DEFINITIONS

### Acute Toxicity

#### a. Acute Toxicity (TUa)

Expressed in Toxic Units Acute (TUa)

$$TUa = \frac{100}{\frac{96\text{-hr LC}}{50\%}}$$

#### b. Lethal Concentration 50% (LC 50)

LC 50 (percent waste giving 50% survival of test organisms) shall be determined by static or continuous flow bioassay techniques using standard marine test species as specified in Ocean Plan Appendix III. If specific identifiable substances in wastewater can be demonstrated by the discharger as being rapidly rendered harmless upon discharge to the marine environment, but not as a result of dilution, the LC 50 may be determined after the test samples are adjusted to remove the influence of those substances.

When it is not possible to measure the 96-hour LC 50 due to greater than 50 percent survival of the test species in 100 percent waste, the toxicity concentration shall be calculated by the expression:

$$TUa = \frac{\log (100 - S)}{1.7}$$

where:

S = percentage survival in 100% waste. If S > 99, TUa shall be reported as zero.

**Areas of Special Biological Significance (ASBS):** Those areas designated by the State Water Board as ocean areas requiring protection of species or biological communities to the extent that alteration of natural water quality is undesirable. All Areas of Special Biological Significance are also classified as a subset of STATE WATER QUALITY PROTECTION AREAS.

**Arithmetic Mean ( $\mu$ ),** also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

$$\text{Arithmetic mean} = \mu = \Sigma x / n$$

where:  $\Sigma x$  is the sum of the measured ambient water concentrations, and n is the number of samples.

**Average Monthly Effluent Limitation (AMEL):** the highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

**Average Weekly Effluent Limitation (AWEL):** the highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

**Bioaccumulative** pollutants are those substances taken up by an organism from its surrounding medium through gill membranes, epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

**Carcinogenic** pollutants are substances that are known to cause cancer in living organisms.

**Chlordane** shall mean the sum of chlordane-alpha, chlordane-gamma, chlordene-alpha, chlordene-gamma, nonachlor-alpha, nonachlor-gamma, and oxychlordane.

**Chronic Toxicity:** This parameter shall be used to measure the acceptability of waters for supporting a healthy marine biota until improved methods are developed to evaluate biological response.

a. Chronic Toxicity (TU<sub>c</sub>)

Expressed as Toxic Units Chronic (TU<sub>c</sub>)

$$TU_c = \frac{100}{NOEL}$$

b. No Observed Effect Level (NOEL)

The NOEL is expressed as the maximum percent effluent or receiving water that causes no observable effect on a test organism, as determined by the result of a critical life stage toxicity test listed in Ocean Plan Appendix III, Table III-1.

**Coefficient of Variation (CV):** the measure of the data variability that is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

**Daily Discharge:** Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

**DDT** shall mean the sum of 4,4'DDT, 2,4'DDT, 4,4'DDE, 2,4'DDE, 4,4'DDD, and 2,4'DDD.

**Degrade:** Degradation shall be determined by comparison of the waste field and reference site(s) for characteristic species diversity, population density, contamination, growth anomalies, debility, or supplanting of normal species by undesirable plant and animal species. Degradation occurs if there are significant differences in any of three major biotic groups, namely, demersal fish, benthic invertebrates, or attached algae. Other groups may be evaluated where benthic species are not affected, or are not the only ones affected.

**Detected, but Not Quantified (DNQ):** sample results less than the reported Minimum Level, but greater than or equal to the laboratory's MDL.

**Dichlorobenzenes** shall mean the sum of 1,2- and 1,3-dichlorobenzene.

**Downstream Ocean Waters:** Waters downstream with respect to ocean currents.

**Dredged Material:** Any material excavated or dredged from the navigable waters of the United States, including material otherwise referred to as "spoil".

**Effective Concentration (EC)** is a point estimate of the toxicant concentration that would cause an adverse effect on a quantal, "all or nothing," response (such as death, immobilization, or serious incapacitation) in a given percent of the test organisms. If the effect is death or immobility, the term lethal concentration (LC) may be used. EC values may be calculated using point estimation techniques such as probit, logit, and Spearman-Kärber. EC25 is the concentration of toxicant (in percent effluent) that causes a response in 25 percent of the test organisms.

**Enclosed Bays** means indentations along the coast that enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between the headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. Enclosed bays include, but are not limited to, Humboldt Bay, Bodega Harbor, Tomales Bay, Drake's Estero, San Francisco Bay, Morro Bay, Los Angeles-Long Beach Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay. Enclosed bays do not include inland surface waters or ocean waters.

**Endosulfan:** The sum of endosulfan-alpha and -beta and endosulfan sulfate.

**Estimated Chemical Concentration** is the estimated chemical concentration that results from the confirmed detection of the substance by the analytical method below the ML value.

**Estuaries and Coastal Lagoons** are waters at the mouths of streams that serve as areas of mixing for fresh and ocean waters during a major portion of the year. Mouths of streams that are temporarily separated from the ocean by sandbars shall be considered estuaries. Estuarine waters shall be considered to extend from a bay or the open ocean to the upstream limit of tidal action but may be considered to extend seaward if significant mixing of fresh and salt water occurs in the open coastal waters. The waters described by this definition include but are not limited to the Sacramento-San Joaquin Delta, as defined by Section 12220 of the California Water Code, Suisun Bay, Carquinez Strait downstream to the Carquinez Bridge, and appropriate areas of the Smith, Klamath, Mad, Eel, Noyo, and Russian Rivers.

**Halomethanes** shall mean the sum of bromoform, bromomethane (methyl bromide) and chloromethane (methyl chloride).

**HCH** shall mean the sum of the alpha, beta, gamma (lindane) and delta isomers of hexachlorocyclohexane.

**Inhibition Concentration (IC)** is a point estimate of the toxicant concentration that would cause a given percent reduction in a non-lethal, non-quantal biological measurement, such as growth. For example, an IC25 is the estimated concentration of toxicant that would cause a 25 percent reduction in average young per female or growth. IC values may be calculated using a linear interpolation method such as U.S. EPA's Bootstrap Procedure.

**Initial Dilution** is the process that results in the rapid and irreversible turbulent mixing of wastewater with ocean water around the point of discharge.

For a submerged buoyant discharge, characteristic of most municipal and industrial wastes that are released from the submarine outfalls, the momentum of the discharge and its initial buoyancy act together to produce turbulent mixing. Initial dilution in this case is completed when the diluting wastewater ceases to rise in the water column and first begins to spread horizontally.

For shallow water submerged discharges, surface discharges, and non-buoyant discharges, characteristic of cooling water wastes and some individual discharges, turbulent mixing results primarily from the momentum of discharge. Initial dilution, in these cases, is considered to be completed when the momentum induced velocity of the discharge ceases to produce significant mixing of the waste, or the diluting plume reaches a fixed distance from the discharge to be specified by the Regional Board, whichever results in the lower estimate for initial dilution.

**Instantaneous Maximum Effluent Limitation:** the highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

**Instantaneous Minimum Effluent Limitation:** the lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

**Kelp Beds**, for or purposes of the bacteriological standards of the Ocean Plan, are significant aggregations of marine algae of the genera *Macrocystis* and *Nereocystis*. Kelp beds include the total foliage canopy of *Macrocystis* and *Nereocystis* throughout the water column.

**Lowest Observed Effect Concentration (LOEC)** is the lowest concentration of toxicant to which organisms are exposed in a test, which causes statistically significant adverse effects on the test organisms (i.e., where the values for the observed endpoints are statistically significantly different from the control).

**Mariculture:** The culture of plants and animals in marine waters independent of any pollution source.

**Material:** (a) In common usage: (1) the substance or substances of which a thing is made or composed (2) substantial; (b) For purposes of the Ocean Plan relating to waste disposal, dredging and the disposal of dredged material and fill, MATERIAL means matter of any kind or description which is subject to regulation as waste, or any material dredged from the navigable waters of the United States. See also, DREDGED MATERIAL.

**Maximum Daily Effluent Limitation (MDEL)** means the highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

**Median** is the middle measurement in a set of data. The median of a set of data is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of measurements ( $n$ ) is odd, then the median =  $X_{(n+1)/2}$ . If  $n$  is even, then the median =  $(X_{n/2} + X_{(n/2)+1})/2$  (i.e., the midpoint between the  $n/2$  and  $n/2+1$ ).

**Method Detection Limit (MDL)** is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analyte concentration is greater than zero, as defined in title 40 of the Code of Federal Regulations, Part 136, Attachment B, revised as of July 3, 1999.

**Minimum Level (ML)** is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

**Natural Light:** Reduction of natural light may be determined by the Regional Water Board by measurement of light transmissivity or total irradiance, or both, according to the monitoring needs of the Regional Water Board.

**No Observed Effect Concentration (NOEC)** is the highest tested concentration of an effluent or a toxicant at which no adverse effects are observed on the aquatic test organisms at a specific time of observation. It is determined using hypothesis testing.

**Not Detected (ND):** those sample results less than the laboratory's MDL.

**Ocean Waters** are the territorial marine waters of the State as defined by California law to the extent these waters are outside of enclosed bays, estuaries, and coastal lagoons. Discharges to ocean waters are regulated in accordance with the State Water Board's California Ocean Plan.

**PAHs (polynuclear aromatic hydrocarbons)** shall mean the sum of acenaphthylene, anthracene, 1,2-benzanthracene, 3,4-benzofluoranthene, benzo[k]fluoranthene, 1,12-benzoperylene, benzo[a]pyrene, chrysene, dibenzo[ah]anthracene, fluorene, indeno[1,2,3-cd]pyrene, phenanthrene and pyrene.

**PCBs (polychlorinated biphenyls)** shall mean the sum of chlorinated biphenyls whose analytical characteristics resemble those of Aroclor-1016, Aroclor-1221, Aroclor-1232, Aroclor-1242, Aroclor-1248, Aroclor-1254 and Aroclor-1260.

**Pollutant Minimization Program (PMP)** means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Regional Water Board may consider cost effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan, if required pursuant to Water Code section 13263.3(d), shall be considered to fulfill the PMP requirements.

**Pollution Prevention** means any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in Water Code section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the State or Regional Water Board.

**Recycled Water:** Water which, as a result of treatment of municipal wastewater, is suitable for a direct beneficial use or a controlled use that would not otherwise occur and is therefore considered a valuable resource (Water Code section 13050). The terms "recycled water" and "reclaimed water" have the same meaning (Water Code section 26).

**Reported Minimum Level:** The ML (and its associated analytical method) chosen by the Discharger for reporting and compliance determination from the MLs included in this Order. The MLs included in this Order correspond to approved analytical methods for reporting a sample result that are selected by the Regional Water Board either from Appendix II of the Ocean Plan in accordance with section III.C.5.a. of the Ocean Plan or established in accordance with section III.C.5.b. of the Ocean Plan. The ML is based on the proper application of method-based analytical procedures for sample preparation and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied in cases where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the RL.

**Satellite Collection System** is the portion, if any, of a sanitary sewer system owned or operated by a different public agency than the agency that owns and operates the wastewater treatment facility that a sanitary sewer system is tributary to.

**Shellfish:** Organisms identified by the California Department of Public Health as shellfish for public health purposes (i.e., mussels, clams and oysters).

**Significant Difference:** Defined as a statistically significant difference in the means of two distributions of sampling results at the 95 percent confidence level.

**Six-Month Median Effluent Limitation:** The highest allowable moving median of all daily discharges for any 180-day period.

**Source of Drinking Water** is any water designated as municipal or domestic supply (MUN) in a Regional Water Board Basin Plan.

**Standard Deviation ( $\sigma$ )** is a measure of variability that is calculated as follows:

$$\sigma = \left( \frac{\sum[(x - \mu)^2]}{(n - 1)} \right)^{0.5}$$

where:

x is the observed value;

$\mu$  is the arithmetic mean of the observed values; and

n is the number of samples.

**State Water Quality Protection Areas (SWQPAs):** Non-terrestrial marine or estuarine areas designated to protect marine species or biological communities from an undesirable alteration in natural water quality. All AREAS OF SPECIAL BIOLOGICAL SIGNIFICANCE (ASBS) that

were previously designated by the State Water Board in Resolution No. 74-28, 74-32, and 75-61 are now also classified as a subset of State Water Quality Protection Areas and require special protections afforded by the Ocean Plan.

**TCDD Equivalents** shall mean the sum of the concentrations of chlorinated dibenzodioxins (2,3,7,8-CDDs) and chlorinated dibenzofurans (2,3,7,8-CDFs) multiplied by their respective toxicity factors, as shown in the table below.

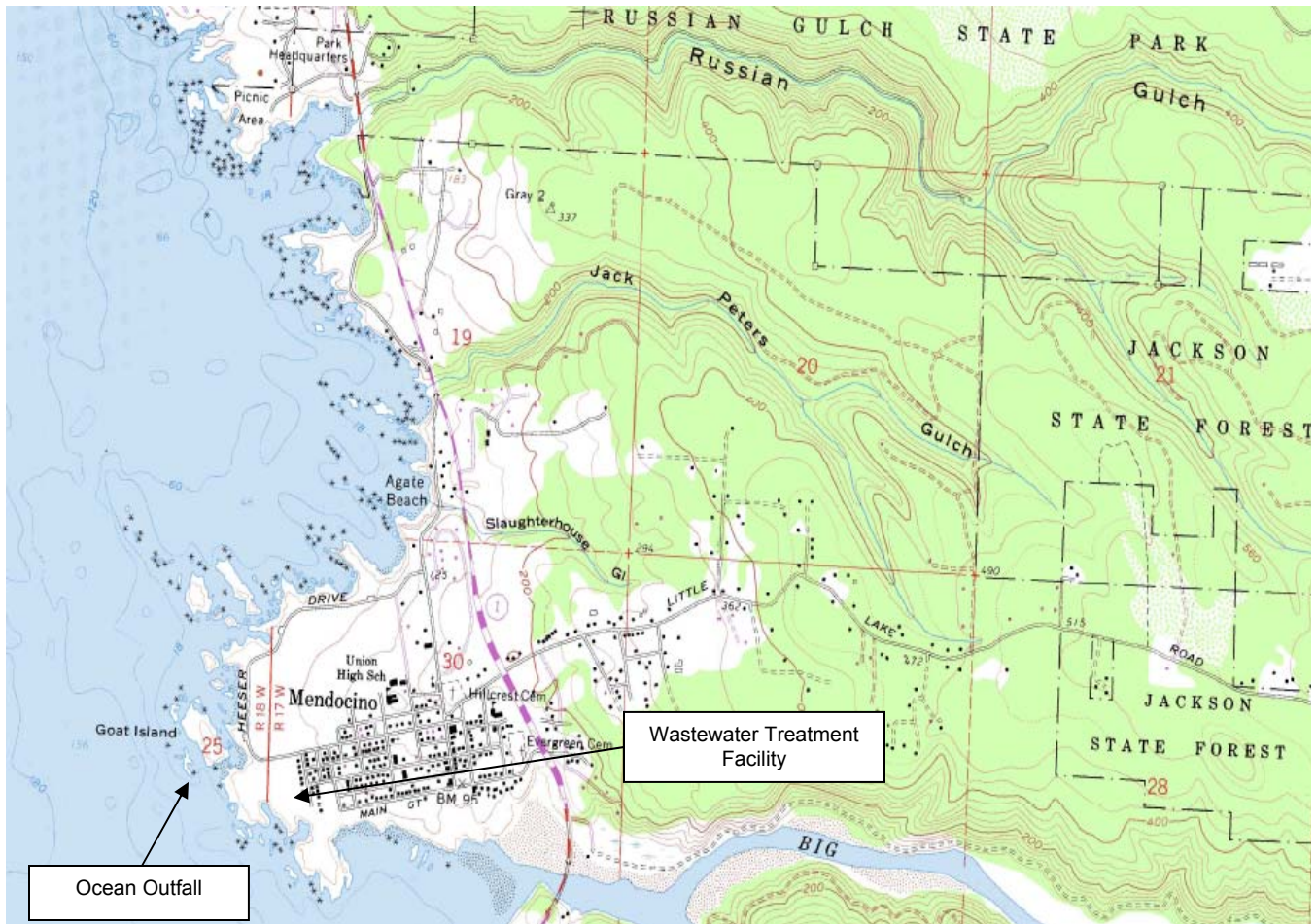
| Isomer Group        | Toxicity<br>Equivalence Factor |
|---------------------|--------------------------------|
| 2,3,7,8-tetra CDD   | 1.0                            |
| 2,3,7,8-penta CDD   | 0.5                            |
| 2,3,7,8-hexa CDDs   | 0.1                            |
| 2,3,7,8-hepta CDD   | 0.01                           |
| octa CDD            | 0.001                          |
| 2,3,7,8 tetra CDF   | 0.1                            |
| 1,2,3,7,8 penta CDF | 0.05                           |
| 2,3,4,7,8 penta CDF | 0.5                            |
| 2,3,7,8 hexa CDFs   | 0.1                            |
| 2,3,7,8 hepta CDFs  | 0.01                           |
| octa CDF            | 0.001                          |

**Toxicity Reduction Evaluation (TRE)** is a study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. (A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.)

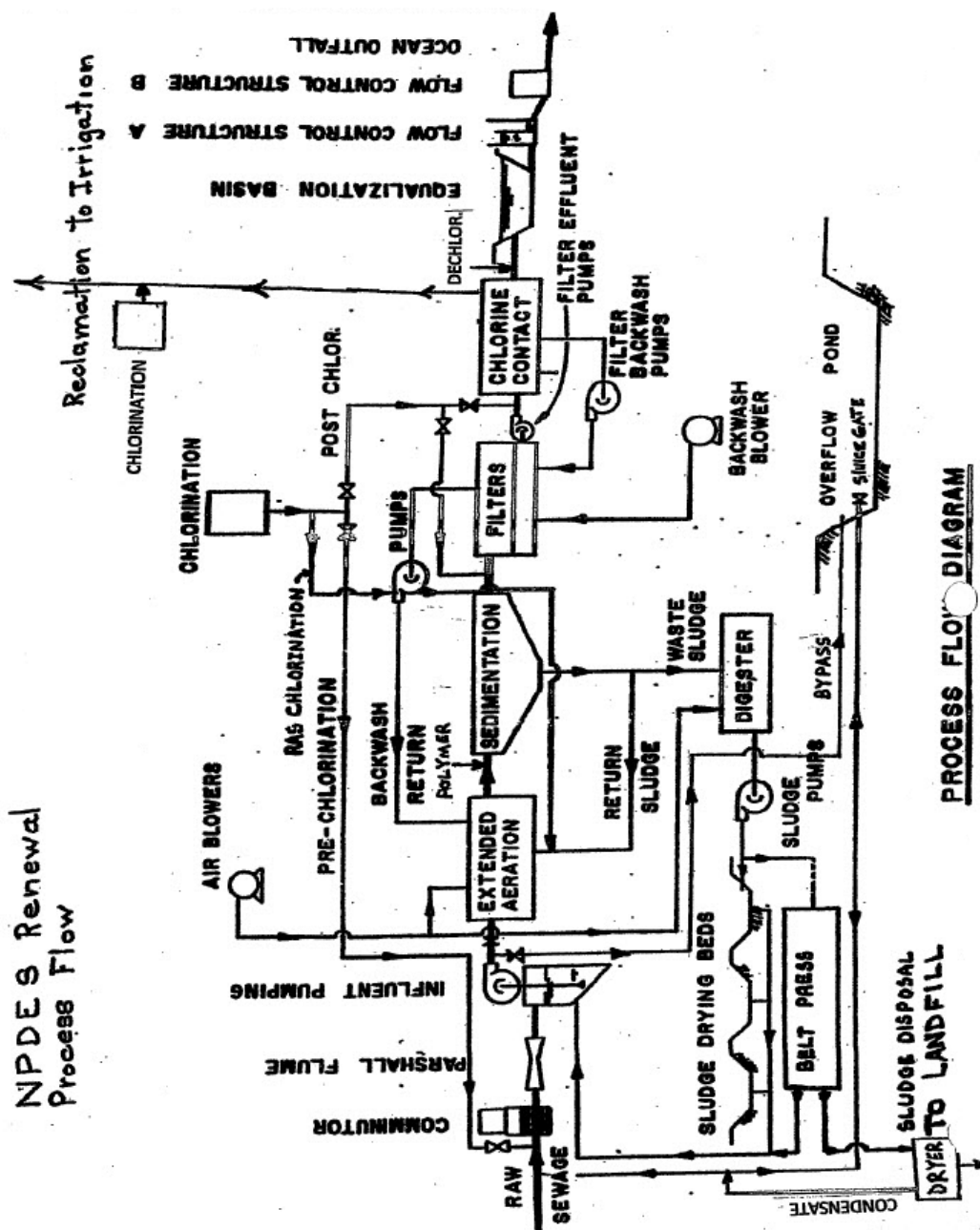
**Waste:** As used in the Ocean Plan, waste includes a Discharger's total discharge, of whatever origin, i.e., gross, not net, discharge.

**Water Reclamation:** The treatment of wastewater to render it suitable for reuse, the transportation of treated wastewater to the place of use, and the actual use of treated wastewater for a direct beneficial use or controlled use that would not otherwise occur.

## ATTACHMENT B – MAP



# ATTACHMENT C – FLOW SCHEMATIC



## **ATTACHMENT D –STANDARD PROVISIONS**

### **I. STANDARD PROVISIONS – PERMIT COMPLIANCE**

#### **A. Duty to Comply**

1. The Discharger must comply with all of the conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA) and the California Water Code and is grounds for enforcement action, for permit termination, revocation and reissuance, or modification; or denial of a permit renewal application. (40 C.F.R. § 122.41(a).)
2. The Discharger shall comply with effluent standards or prohibitions established under Section 307(a) of the CWA for toxic pollutants and with standards for sewage sludge use or disposal established under Section 405(d) of the CWA within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement. (40 C.F.R. § 122.41(a)(1).)

#### **B. Need to Halt or Reduce Activity Not a Defense**

It shall not be a defense for a Discharger in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order. (40 C.F.R. § 122.41(c).)

#### **C. Duty to Mitigate**

The Discharger shall take all reasonable steps to minimize or prevent any discharge or sludge use or disposal in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment. (40 C.F.R. § 122.41(d).)

#### **D. Proper Operation and Maintenance**

The Discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Discharger to achieve compliance with the conditions of this Order. Proper operation and maintenance also includes adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Discharger only when necessary to achieve compliance with the conditions of this Order. (40 C.F.R. § 122.41(e).)

#### **E. Property Rights**

1. This Order does not convey any property rights of any sort or any exclusive privileges. (40 C.F.R. § 122.41(g).)

2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of state or local law or regulations. (40 C.F.R. § 122.5(c).)

#### **F. Inspection and Entry**

The Discharger shall allow the Regional Water Board, State Water Board, United States Environmental Protection Agency (USEPA), and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to (40 C.F.R. § 122.41(i); Wat. Code, § 13383):

1. Enter upon the Discharger's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order (40 C.F.R. § 122.41(i)(1));
2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order (40 C.F.R. § 122.41(i)(2));
3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order (40 C.F.R. § 122.41(i)(3)); and
4. Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the Water Code, any substances or parameters at any location. (40 C.F.R. § 122.41(i)(4).)

#### **G. Bypass**

1. Definitions
  - a. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility. (40 C.F.R. § 122.41(m)(1)(i).)
  - b. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 C.F.R. § 122.41(m)(1)(ii).)
2. Bypass not exceeding limitations. The Discharger may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance I.G.3, I.G.4, and I.G.5 below. (40 C.F.R. § 122.41(m)(2).)

3. Prohibition of bypass. Bypass is prohibited, and the Regional Water Board may take enforcement action against a Discharger for bypass, unless (40 C.F.R. § 122.41(m)(4)(i)):
  - a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 C.F.R. § 122.41(m)(4)(i)(A));
  - b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 C.F.R. § 122.41(m)(4)(i)(B)); and
  - c. The Discharger submitted notice to the Regional Water Board as required under Standard Provisions – Permit Compliance I.G.5 below. (40 C.F.R. § 122.41(m)(4)(i)(C).)
4. The Regional Water Board may approve an anticipated bypass, after considering its adverse effects, if the Regional Water Board determines that it will meet the three conditions listed in Standard Provisions – Permit Compliance I.G.3 above. (40 C.F.R. § 122.41(m)(4)(ii).)
5. Notice
  - a. Anticipated bypass. If the Discharger knows in advance of the need for a bypass, it shall submit a notice, if possible at least 10 days before the date of the bypass. (40 C.F.R. § 122.41(m)(3)(i).)
  - b. Unanticipated bypass. The Discharger shall submit notice of an unanticipated bypass as required in Standard Provisions - Reporting V.E below (24-hour notice). (40 C.F.R. § 122.41(m)(3)(ii).)

## **H. Upset**

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology based permit effluent limitations because of factors beyond the reasonable control of the Discharger. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation. (40 C.F.R. § 122.41(n)(1).)

1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance I.H.2 below are met. No

determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 C.F.R. § 122.41(n)(2).).

2. Conditions necessary for a demonstration of upset. A Discharger who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that (40 C.F.R. § 122.41(n)(3)):
  - a. An upset occurred and that the Discharger can identify the cause(s) of the upset (40 C.F.R. § 122.41(n)(3)(i));
  - b. The permitted facility was, at the time, being properly operated (40 C.F.R. § 122.41(n)(3)(ii));
  - c. The Discharger submitted notice of the upset as required in Standard Provisions – Reporting V.E.2.b below (24-hour notice) (40 C.F.R. § 122.41(n)(3)(iii)); and
  - d. The Discharger complied with any remedial measures required under Standard Provisions – Permit Compliance I.C above. (40 C.F.R. § 122.41(n)(3)(iv).)
3. Burden of proof. In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof. (40 C.F.R. § 122.41(n)(4).)

## **II. STANDARD PROVISIONS – PERMIT ACTION**

### **A. General**

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Discharger for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. (40 C.F.R. § 122.41(f).)

### **B. Duty to Reapply**

If the Discharger wishes to continue an activity regulated by this Order after the expiration date of this Order, the Discharger must apply for and obtain a new permit. (40 C.F.R. § 122.41(b).)

### **C. Transfers**

This Order is not transferable to any person except after notice to the Regional Water Board. The Regional Water Board may require modification or revocation and reissuance of the Order to change the name of the Discharger and incorporate such other

requirements as may be necessary under the CWA and the Water Code. (40 C.F.R. § 122.41(l)(3); § 122.61.)

### **III. STANDARD PROVISIONS – MONITORING**

- A.** Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 C.F.R. § 122.41(j)(1).)
- B.** Monitoring results must be conducted according to test procedures under Part 136 or, in the case of sludge use or disposal, approved under Part 136 unless otherwise specified in Part 503 unless other test procedures have been specified in this Order. (40 C.F.R. § 122.41(j)(4); § 122.44(i)(1)(iv).)

### **IV. STANDARD PROVISIONS – RECORDS**

- A.** Except for records of monitoring information required by this Order related to the Discharger's sewage sludge use and disposal activities, which shall be retained for a period of at least five years (or longer as required by Part 503), the Discharger shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the Regional Water Board Executive Officer at any time. (40 C.F.R. § 122.41(j)(2).)
- B.** Records of monitoring information shall include:
  - 1. The date, exact place, and time of sampling or measurements (40 C.F.R. § 122.41(j)(3)(i));
  - 2. The individual(s) who performed the sampling or measurements (40 C.F.R. § 122.41(j)(3)(ii));
  - 3. The date(s) analyses were performed (40 C.F.R. § 122.41(j)(3)(iii));
  - 4. The individual(s) who performed the analyses (40 C.F.R. § 122.41(j)(3)(iv));
  - 5. The analytical techniques or methods used (40 C.F.R. § 122.41(j)(3)(v)); and
  - 6. The results of such analyses. (40 C.F.R. § 122.41(j)(3)(vi).)
- C.** Claims of confidentiality for the following information will be denied (40 C.F.R. § 122.7(b)):
  - 1. The name and address of any permit applicant or Discharger (40 C.F.R. § 122.7(b)(1)); and

2. Permit applications and attachments, permits and effluent data. (40 C.F.R. § 122.7(b)(2).)

## **V. STANDARD PROVISIONS – REPORTING**

### **A. Duty to Provide Information**

The Discharger shall furnish to the Regional Water Board, State Water Board, or USEPA within a reasonable time, any information which the Regional Water Board, State Water Board, or USEPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine compliance with this Order. Upon request, the Discharger shall also furnish to the Regional Water Board, State Water Board, or USEPA copies of records required to be kept by this Order. (40 C.F.R. § 122.41(h); Wat. Code, § 13267.)

### **B. Signatory and Certification Requirements**

1. All applications, reports, or information submitted to the Regional Water Board, State Water Board, and/or USEPA shall be signed and certified in accordance with Standard Provisions – Reporting V.B.2, V.B.3, V.B.4, and V.B.5 below. (40 C.F.R. § 122.41(k).)
2. All permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of USEPA). (40 C.F.R. § 122.22(a)(3).)
3. All reports required by this Order and other information requested by the Regional Water Board, State Water Board, or USEPA shall be signed by a person described in Standard Provisions – Reporting V.B.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
  - a. The authorization is made in writing by a person described in Standard Provisions – Reporting V.B.2 above (40 C.F.R. § 122.22(b)(1));
  - b. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 C.F.R. § 122.22(b)(2)); and

- c. The written authorization is submitted to the Regional Water Board and State Water Board. (40 C.F.R. § 122.22(b)(3).)
4. If an authorization under Standard Provisions – Reporting V.B.3 above is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting V.B.3 above must be submitted to the Regional Water Board and State Water Board prior to or together with any reports, information, or applications, to be signed by an authorized representative. (40 C.F.R. § 122.22(c).)
5. Any person signing a document under Standard Provisions – Reporting V.B.2 or V.B.3 above shall make the following certification:  
  
“I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.” (40 C.F.R. § 122.22(d).)

### **C. Monitoring Reports**

1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order. (40 C.F.R. § 122.22(l)(4).)
2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Regional Water Board or State Water Board for reporting results of monitoring of sludge use or disposal practices. (40 C.F.R. § 122.41(l)(4)(i).)
3. If the Discharger monitors any pollutant more frequently than required by this Order using test procedures approved under Part 136 or, in the case of sludge use or disposal, approved under Part 136 unless otherwise specified in Part 503, or as specified in this Order, the results of this monitoring shall be included in the calculation and reporting of the data submitted in the DMR or sludge reporting form specified by the Regional Water Board. (40 C.F.R. § 122.41(l)(4)(ii).)
4. Calculations for all limitations, which require averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order. (40 C.F.R. § 122.41(l)(4)(iii).)

#### **D. Compliance Schedules**

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date. (40 C.F.R. § 122.41(l)(5).)

#### **E. Twenty-Four Hour Reporting**

1. The Discharger shall report any noncompliance that may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A written submission shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The written submission shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance. (40 C.F.R. § 122.41(l)(6)(i).)
2. The following shall be included as information that must be reported within 24 hours under this paragraph (40 C.F.R. § 122.41(l)(6)(ii)):
  - a. Any unanticipated bypass that exceeds any effluent limitation in this Order. (40 C.F.R. § 122.41(l)(6)(ii)(A).)
  - b. Any upset that exceeds any effluent limitation in this Order. (40 C.F.R. § 122.41(l)(6)(ii)(B).)
3. The Regional Water Board may waive the above-required written report under this provision on a case-by-case basis if an oral report has been received within 24 hours. (40 C.F.R. § 122.41(l)(6)(iii).)

#### **F. Planned Changes**

The Discharger shall give notice to the Regional Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when (40 C.F.R. § 122.41(l)(1)):

1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in section 122.29(b) (40 C.F.R. § 122.41(l)(1)(i)); or
2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are not subject to effluent limitations in this Order. (40 C.F.R. § 122.41(l)(1)(ii).)
3. The alteration or addition results in a significant change in the Discharger's sludge use or disposal practices, and such alteration, addition, or change may justify the

application of permit conditions that are different from or absent in the existing permit, including notification of additional use or disposal sites not reported during the permit application process or not reported pursuant to an approved land application plan. (40 C.F.R. § 122.41(l)(1)(iii).)

#### **G. Anticipated Noncompliance**

The Discharger shall give advance notice to the Regional Water Board or State Water Board of any planned changes in the permitted facility or activity that may result in noncompliance with General Order requirements. (40 C.F.R. § 122.41(l)(2).)

#### **H. Other Noncompliance**

The Discharger shall report all instances of noncompliance not reported under Standard Provisions – Reporting V.C, V.D, and V.E above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting V.E above. (40 C.F.R. § 122.41(l)(7).)

#### **I. Other Information**

When the Discharger becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Regional Water Board, State Water Board, or USEPA, the Discharger shall promptly submit such facts or information. (40 C.F.R. § 122.41(l)(8).)

### **VI. STANDARD PROVISIONS – ENFORCEMENT**

- A.** The Regional Water Board is authorized to enforce the terms of this permit under several provisions of the Water Code, including, but not limited to, sections 13385, 13386, and 13387.

### **VII. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS**

#### **A. Publicly-Owned Treatment Works (POTWs)**

All POTWs shall provide adequate notice to the Regional Water Board of the following (40 C.F.R. § 122.42(b)):

1. Any new introduction of pollutants into the POTW from an indirect discharger that would be subject to sections 301 or 306 of the CWA if it were directly discharging those pollutants (40 C.F.R. § 122.42(b)(1)); and
2. Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of the Order. (40 C.F.R. § 122.42(b)(2).)

3. Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 C.F.R. § 122.42(b)(3).)

## ATTACHMENT E – MONITORING AND REPORTING PROGRAM

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## ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)

The Code of Federal Regulations section 122.48 requires that all NPDES permits specify monitoring and reporting requirements. Water Code sections 13267 and 13383 also authorize the Regional Water Quality Control Board (Regional Water Board) to require technical and monitoring reports. This MRP establishes monitoring and reporting requirements, which implement the federal and California regulations.

### I. GENERAL MONITORING PROVISIONS

- A. Wastewater Monitoring Provision. Composite samples may be taken by a proportional sampling device approved by the Executive Officer or by grab samples composited in proportion to flow. In compositing grab samples, the sampling interval shall not exceed one hour.
- B. If the Discharger monitors any pollutant more frequently than required by this Order, using test procedures approved by section 136 or as specified in this Order, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the monthly and annual discharger monitoring reports.
- C. Laboratories analyzing monitoring samples shall be certified by the Department of Public Health, in accordance with the provisions of Water Code section 13176, and must include quality assurance / quality control data with their analytical reports.

### II. MONITORING LOCATIONS

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order.

**Table E-1. Monitoring Station Locations**

| Discharge Point | Monitoring Location | Monitoring Location Description                                                                                                                              |
|-----------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ---             | INF-001             | Headworks, where representative samples of influent wastewater can be collected and flow measured prior to treatment.                                        |
| 001             | EFF-001             | Effluent Wet Well, immediately following the tertiary filters                                                                                                |
| 001             | EFF-002             | Chlorine Contact Tank, where treated wastewater can be collected prior to discharge to the Flow Equalization Pond.                                           |
| 001             | EFF-003             | Structure A, where overflow from the Flow Equalization Pond can be collected before contact with the receiving water.                                        |
| 002             | REC-001             | Location where samples of treated wastewater can be collected that are representative of water to be reclaimed at the Mendocino High School athletic fields. |

| Discharge Point | Monitoring Location | Monitoring Location Description                                                                             |
|-----------------|---------------------|-------------------------------------------------------------------------------------------------------------|
| ---             | RSW-001             | Location in the receiving water in the vicinity of the outfall diffuser.                                    |
| ---             | RSW-002             | Location in the receiving water outside the influence of the discharge, for determining ambient conditions. |

### III. INFLUENT MONITORING REQUIREMENTS

#### A. Monitoring Location INF-001

1. The Discharger shall monitor influent to the wastewater treatment facility at Monitoring Location Name INF-001 as follows.

**Table E-2. Influent Monitoring**

| Parameter         | Units | Sample Type     | Minimum Sampling Frequency | Required Analytical Method |
|-------------------|-------|-----------------|----------------------------|----------------------------|
| BOD <sub>5</sub>  | mg/L  | 24-hr composite | Weekly <sup>1</sup>        | Standard Methods           |
| TSS               | mg/L  | 24-hr composite | Weekly <sup>1</sup>        | Standard Methods           |
| Flow <sup>2</sup> | mgd   | Continuous      | Continuous                 | Meter                      |

### IV. EFFLUENT MONITORING REQUIREMENTS

#### A. Monitoring Location EFF-001

1. The Discharger shall monitor treated wastewater to be discharged at Discharge Point 001, with compliance determined at Monitoring Location EFF-001, as follows.

**Table E-3. Effluent Monitoring, Monitoring Location EFF-001**

| Parameter        | Units | Sample Type     | Minimum Sampling Frequency | Required Analytical Method |
|------------------|-------|-----------------|----------------------------|----------------------------|
| BOD <sub>5</sub> | mg/L  | 24-hr composite | Weekly                     | SM 5210 B                  |
| TSS              | mg/L  | 24-hr composite | Weekly                     | SM 2540 D                  |

<sup>1</sup> Monitoring of BOD<sub>5</sub> and TSS in influent shall coincide with monitoring of these parameters in effluent. For compliance determination, weekly and monthly averages and medians will be based on the calendar weeks (Sunday through Saturday) and months.

<sup>2</sup> For each month, the Discharger shall report the maximum daily and mean daily influent flow rates.

| Parameter         | Units   | Sample Type     | Minimum Sampling Frequency | Required Analytical Method |
|-------------------|---------|-----------------|----------------------------|----------------------------|
| Settleable Solids | mL/L-hr | 24-hr composite | Weekly                     | SM 2540 F                  |
| Turbidity         | NTU     | Continuous      | Daily                      | 40 CFR 136                 |
| Oil and Grease    | mg/L    | Grab            | Quarterly                  | 40 CFR 136                 |

2. The Discharger shall monitor treated wastewater to be discharged at Discharge Point 001, with compliance determined at Monitoring Location EFF-002, as follows.

**Table E-4. Effluent Monitoring, Monitoring Location EFF-002**

| Parameter               | Units      | Sample Type | Minimum Sampling Frequency | Required Analytical Method |
|-------------------------|------------|-------------|----------------------------|----------------------------|
| Total Coliform Bacteria | MPN/100 mL | Grab        | Weekly                     | 40 CFR 136                 |
| pH                      | s.u.       | Grab        | Weekly                     | 40 CFR 136                 |

3. The Discharger shall monitor treated wastewater to be discharged at Discharge Point 001, with compliance determined at Monitoring Location EFF-003, as follows.

**Table E-5. Effluent Monitoring, Monitoring Location EFF-003**

| Parameter                                            | Units  | Sample Type                 | Minimum Sampling Frequency | Required Analytical Method |
|------------------------------------------------------|--------|-----------------------------|----------------------------|----------------------------|
| Chlorine Residual <sup>3</sup>                       | mg/L   | Grab                        | Weekly                     | 40 CFR 136                 |
| Chronic Toxicity                                     | TUc    | Grab                        | Annually                   | 40 CFR 136                 |
| TCDD Equivalents                                     | µg/L   | 24-hr composite             | Annually                   | 40 CFR 136 <sup>4</sup>    |
| Chloroform                                           | µg/L   | Grab                        | Quarterly                  | 40 CFR 136                 |
| Dichlorobromomethane                                 | µg/L   | Grab                        | Quarterly                  | 40 CFR 136                 |
| Total Ammonia                                        | mg/L N | Grab                        | Monthly                    | 40 CFR 136                 |
| Remaining Ocean Plan Table B Pollutants <sup>5</sup> | µg/L   | Grab/composite <sup>6</sup> | 1/5 Year Permit Term       | 40 CFR 136                 |

<sup>3</sup> The Discharger shall monitor total residual chlorine in the effluent daily at a point following dechlorination using a method with a reporting limit no greater than 0.1 mg/L.

<sup>4</sup> For toxic pollutants with effluent limitations, detection limits shall be below the effluent limitations. If the lowest minimum level (ML) published in Appendix II of the Water Quality Control Plan for Ocean Waters of California (Ocean Plan) is not below the effluent limitation, the detection limit shall be the lowest ML. For toxic pollutants without effluent limitations, the detection limits shall be equal to or less than the lowest ML published in Appendix II of the Ocean Plan.

<sup>5</sup> Those pollutants identified by Table B of the Ocean Plan, excluding acute toxicity and those pollutants with specific monitoring requirements listed in Table E-3.

<sup>6</sup> For volatile constituents, grab samples of at least 100 ML shall be collected over a period not exceeding 15 minutes.

## V. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

### A. Chronic Toxicity Testing

The Discharger shall conduct chronic toxicity testing to demonstrate compliance with the chronic toxicity water quality objective contained in the Ocean Plan. The Discharger shall meet the following chronic toxicity testing requirements:

1. **Test Frequency.** The Discharger shall conduct annual chronic toxicity testing at Discharge Point 001.
2. **Sample Type.** For 96-hour static renewal or 96-hour static non-renewal testing, effluent samples from Monitoring Locations EFF-003 shall be grab samples that are representative of the volume and quality of the discharge from the facility. For toxicity tests requiring renewals, grab samples collected on consecutive days are required.
3. **Test Species.** Critical life stage bioassay testing shall be conducted using an approved test, and test species, as described by Table III-1 of the Ocean Plan and presented below. Initial testing for the first suite of tests, shall be conducted with a vertebrate, an invertebrate, and a plant species, and thereafter, monitoring can be reduced to the most sensitive species. At least once every five years, the Discharger shall rescreen once with the three species listed above, and continue to monitor with the most sensitive species.

**Table E-6. Approved Tests – Chronic Toxicity**

| Species                                                                                   | Test                                         | Tier <sup>1</sup> | Reference <sup>2</sup> |
|-------------------------------------------------------------------------------------------|----------------------------------------------|-------------------|------------------------|
| Giant kelp, <i>Macrocystis pyrifera</i>                                                   | percent germination; germ tube length        | 1                 | a, c                   |
| Red abalone, <i>Haliotis rufescens</i>                                                    | abnormal shell development                   | 1                 | a, c                   |
| Oyster, <i>Crassostrea gigas</i> ; mussels, <i>Mytilus spp.</i>                           | abnormal shell development; percent survival | 1                 | a, c                   |
| Urchin, <i>Strongylocentrotus purpuratus</i> ; sand dollar, <i>Dendraster excentricus</i> | percent normal development                   | 1                 | a, c                   |
| Urchin, <i>Strongylocentrotus purpuratus</i> ; sand dollar, <i>Dendraster excentricus</i> | percent fertilization                        | 1                 | a, c                   |
| Shrimp, <i>Homesimysis costata</i>                                                        | percent survival; growth                     | 1                 | a, c                   |
| Shrimp, <i>Mysidopsis bahia</i>                                                           | percent survival; fecundity                  | 2                 | b, d                   |
| Topsmelt, <i>Atherinops affinis</i>                                                       | larval growth rate; percent survival         | 1                 | a, c                   |
| Silverside, <i>Menidia beryllina</i>                                                      | larval growth rate; percent survival         | 2                 | b, d                   |

---

<sup>1</sup> First tier methods are preferred for compliance monitoring. If first tier organisms are not available, the Discharger can use a second tier test method following approval by the Regional Water Board.

<sup>2</sup> Protocol References:

- a. Chapman, G.A., D.L. Denton, and J.M. Lazorchak. 1995. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms. U.S. EPA Report No. EPA/600/R-95/136.
- b. Klemm, D.J., G.E. Morrison, T.J. Norberg-King, W.J. Peltier, and M.A. Heber. 1994. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Water to Marine and Estuarine Organisms. U.S. EPA Report No. EPA-600-4-91-003.
- c. SWRCB 1996. Procedures Manual for Conducting Toxicity Tests Developed by the Marine Bioassay Project. 96-1WQ.
- d. Weber, C.I., W.B. Horning, I.I., D.J. Klemm, T.W. Nieheisel, P.A. Lewis, E.L. Robinson, J. Menkedick and F. Kessler (eds). 1998. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Marine and Estuarine Organisms. EPA/600/4-87/028. National Information Service, Springfield, VA.

4. **Test Methods.** The presence of chronic toxicity shall be estimated as specified in USEPA's *Short-Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Water to West Coast Marine and Estuarine Organisms* (USEPA Report No. EPA/600/R-95/136, or subsequent editions), *Short-Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Water to Marine and Estuarine Organisms* (USEPA Report No. EPA-821-R-02-014 or subsequent editions), or other methods approved by the Executive Officer.
5. **Test Dilutions.** The chronic toxicity test shall be conducted using a series of at least five dilutions and a control. The series shall consist of the following dilution series: 0.25, 0.5, 0.75, 1.0 and 1.25 percent, and a control. Control and dilution water shall be receiving water collected at an appropriate location beyond the influence of the discharge. Laboratory water may be substituted for receiving water, as described in the USEPA test methods manual, upon approval by the Executive Officer. If the dilution water used is different from the culture water, a second control using culture water shall be used.
6. **Reference Toxicant.** If organisms are not cultured in-house, concurrent testing with a reference toxicant shall be conducted. Where organisms are cultured in-house, monthly reference toxicant testing is sufficient. Reference toxicant tests also shall be conducted using the same test conditions as the effluent toxicity tests (e.g., same test duration, etc).
7. **Test Failure.** If either the reference toxicant test or the chronic toxicity test does not meet all test acceptability criteria, as specified in the test method, the Discharger

shall re-sample and re-test as soon as possible, not to exceed 7 days following notification of test failure.

8. **Notification.** The Discharger shall notify the Regional Water Board in writing 14 days after the receipt of test results, which indicate the exceedance of the effluent limitation for chronic toxicity.
9. **Accelerated Monitoring Requirements.** If the result of any chronic toxicity test exceeds the dilution-adjusted water quality objective of 101 TUC, and the testing meets all test acceptability criteria, the Discharger shall initiate accelerated monitoring. Accelerated monitoring shall consist of four additional effluent samples, one test conducted approximately every week over a four week period. Testing shall commence within 14 days of receipt of the sample results of the exceedance of the chronic toxicity effluent limitation. If the discharge will cease before the additional samples can be collected, the Discharger shall contact the Executive Officer within 21 days with a plan to demonstrate compliance with the chronic toxicity effluent limitation.

The following protocol shall be used for accelerated monitoring and TRE implementation:

- a. If the results of four consecutive accelerated monitoring tests do not exceed the dilution-adjusted chronic toxicity water quality objective, the Discharger may cease accelerated monitoring and resume regular annual chronic toxicity monitoring. However, if there is adequate evidence of a pattern of effluent toxicity, the Regional Water Board's Executive Officer may require that the Discharger initiate a TRE.
- b. If the source(s) of the toxicity is easily identified (i.e. temporary plant upset), the Discharger shall make necessary corrections to the facility and shall continue accelerated monitoring until four (4) consecutive accelerated tests do not exceed the dilution-adjusted chronic toxicity water quality objective. Upon confirmation that the chronic toxicity has been removed, the Discharger may cease accelerated monitoring and resume routine chronic toxicity monitoring.
- c. If the result of any accelerated toxicity test exceeds the dilution-adjusted chronic toxicity water quality objective, the Discharger shall cease accelerated monitoring and initiate a TRE to investigate the cause(s) and identify corrective actions to reduce or eliminate the chronic toxicity. Within thirty (30) days of notification by the laboratory of the test results exceeding the effluent limitation for chronic toxicity during accelerated monitoring, the Discharger shall submit a TRE Action Plan to the Regional Water Board including, at minimum:
  - (1) Specific actions the Discharger will take to investigate and identify the cause(s) of toxicity, including a TRE WET monitoring schedule;

(2) Specific actions the Discharger will take to mitigate the impact of the discharge and prevent the recurrence of toxicity; and

(3) A schedule for these actions.

## **B. Chronic Toxicity Reporting**

1. **Routine Reporting.** All toxicity test reports shall include the contracting laboratory's complete report provided to the Discharger and shall be in accordance with the appropriate "Report Preparation and Test Review" sections of the method manuals.

The WET test report shall contain a narrative report that includes details about WET test procedures and results, including the following:

### **a. Test Procedures**

- i. receipt and handling of the effluent sample that includes a tabular summary of initial water quality characteristics;
- ii. the source and make-up of the lab control/diluent water used for the test;
- iii. any manipulations done to lab control/diluent and effluent such as filtration, nutrient addition, etc.;
- iv. identification of any reference toxicant testing performed;
- v. tabular summary of test results for control water and each effluent dilution and statistics summary to include calculation of NO
- vi. EC, TU<sub>c</sub> and IC<sub>25</sub>;
- vii. identification of any anomalies or nuances in the test procedures or results;
- viii. Summary and Conclusions section.

### **b. Test Results. Test results shall include, at a minimum, for each test:**

- i. sample date(s)
- ii. test initiation date
- iii. test species
- iv. end point values for each dilution (e.g., number of young, growth rate, percent survival)
- v. NOEC value(s) in percent effluent
- vi. IC15, IC25, IC40, and IC50 values (or EC15, EC25...etc.) in percent effluent
- vii. TU<sub>c</sub> values (100/NOEC)
- viii. Mean percent mortality ( $\pm$ s.d.) after 96 hours in 100 percent effluent (if applicable)

- ix. NOEC and LOEC values for reference toxicant test(s)
  - x. IC50 or EC50 value(s) for reference toxicant test(s)
  - xi. Available water quality measurements for each test (e.g., pH, DO, temperature, conductivity, hardness, salinity, ammonia)
  - xii. Statistical methods used to calculate endpoints.
  - xiii. The statistical output page, which includes the calculation of percent minimum significant difference (PMSD.)
  - xiv. Results of applicable reference toxicant data with the statistical output page giving the species, NOEC, LOEC, type of toxicant, dilution water used, concentrations used, PMSD and dates tested; the reference toxicant control charts for each endpoint, which include summaries of reference toxicant tests performed by the contracting laboratory; and any information on deviations from standard test procedures or problems encountered in completing the test and how the problems were resolved.
2. **Quality Assurance Reporting.** Because the permit requires sublethal hypothesis testing endpoints from methods 1006.0 and 1007.0 in the test methods manual titled *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Marine and Estuarine Organisms* (EPA-821-R-02-014, 2002), in-test variability must be reviewed for acceptability and variability criteria (upper and lower PMSD bounds) must be applied, as directed under section 10.2.8 – *Test Variability* of the test methods manual. Under section 10.2.8, the calculated PMSD for both reference toxicant test and effluent toxicity test results must be compared with the upper and lower PMSD bounds variability criteria specified in Table 6 – *Variability Criteria (Upper and Lower PMSD Bounds) for Sublethal Hypothesis Testing Endpoints Submitted Under NPDES Permits*, following the review criteria in paragraphs 10.2.8.2.4.1 through 10.2.8.2.4.5 of the test methods manual. Based on this review, only accepted effluent toxicity test results shall be reported.
3. **Compliance Summary:** The results of the chronic toxicity testing shall be provided in the most recent self-monitoring report and shall include a summary table organized by test species, type of test (survival, growth or reproduction) and monitoring frequency of toxicity data from at least three of the most recent samples. The final report shall clearly demonstrate that the Discharger is in compliance with effluent limitations and other permit requirements.

## VI. LAND DISCHARGE MONITORING REQUIREMENTS

Not Applicable.

## VII. RECLAMATION MONITORING REQUIREMENTS

### A. Monitoring Location REC-001

1. The Discharger shall monitor tertiary treated effluent to be reclaimed and used for irrigation at Monitoring Location REC-001 as follows:

**Table E-7. Reclamation Monitoring Requirements - Monitoring Location REC-001**

| Parameter               | Units     | Sample Type | Minimum Sampling Frequency |
|-------------------------|-----------|-------------|----------------------------|
| Total Chlorine Residual | mg/L      | meter       | Daily                      |
| Contact Time            | minutes   | Measurement | Daily                      |
| CT Value                | mg-min./L | Calculation | Daily                      |
| Nitrate (as N)          | mg/L      | Grab        | Monthly                    |

## VIII. RECEIVING WATER MONITORING REQUIREMENTS – SURFACE WATER AND GROUNDWATER

### A. Monitoring Location REF-001

1. **Biological Survey.** The Discharger shall conduct a comparative evaluation of indigenous biota in the vicinity of the outfall using a qualified aquatic biologist, at least once every 5 years. The biologist shall prepare a report of observations, including objectionable aquatic growths, floating particulates or grease and oil, aesthetically undesirable discoloration of the ocean surface, color of fish or shellfish, and any evidence of degradation of indigenous biota attributable to the rate of deposition of inert solids, settleable material, nutrient materials, increased concentrations of organic materials, or increased concentrations of Ocean Plan Table B substances. The Discharger shall submit to the Regional Water Board Executive Officer for approval a Biological Survey work plan no later than **May 1, 2011** in order to complete the survey and prepare a final report by the due date for receipt of an application for permit renewal. The final report shall be submitted no later than **December 12, 2014**.

## IX. OTHER MONITORING REQUIREMENTS

### A. Filtration Process Monitoring

1. **Surface Loading Rate Monitoring (Monitoring Location INT-001)**
  - a. **Monitoring.** The Discharger shall use the influent wastewater flow at INF-001 to determine flow to calculate the daily minimum surface loading rate (in gallons per minute per square foot) for each tertiary filter.

- b. **Compliance.** Compliance with the minimum filter surface loading rate as specified in the State of California Division of Drinking Water and Environmental Management Treatment Technology Report for Recycled Water (September 2008 and future revisions thereto) shall be calculated based on the flow rate through each filter unit.
- c. **Reporting.** The minimum filter daily surface loading rate shall be reported on the monthly self-monitoring report.

2. **Additional Effluent Filter Monitoring (Monitoring Location EFF-001)**

- a. **Monitoring.** The turbidity of the filter effluent shall be continuously measured and recorded. Should the turbidity meter and recorder fail, grab sampling at a minimum frequency of 1.2 hours may be substituted for a period of up to 24 hours. The recorded data shall be maintained by the Discharger for at least three years. The daily average and daily maximum turbidity results shall be reported on the monthly monitoring reports.
- b. **Compliance.** Compliance with the daily average effluent turbidity limitation specified in the California Code of Regulations Water Recycling Criteria, as referenced in section IV.D.1.b. of the Order, shall be determined by averaging all turbidity readings collected in a calendar day. Compliance shall be determined using the levels of recorded turbidity taken at intervals of no more than 1.2 hours over a 24-hour period.
- c. **Reporting.** If the filter effluent turbidity exceeds 2 NTU based on a daily average or if the influent turbidity exceeds 5 NTU for more than 15 minutes, the incident shall be reported within the monthly self-monitoring report. If the filter effluent turbidity exceeds 10 NTU at any time, the flow to the recycled water storage tank shall be ceased immediately and the incident reported to the Regional Water Board in writing with the monthly self-monitoring report.

B. **Ocean Outfall Inspection**

- 1. **Outfall Inspection.** Divers shall visually inspect the outfall structure, including the diffuser ports, at least once during the life of this permit to verify operational status of the outfall. A report documenting outfall condition and maintenance, including any observed cracks, breaks, malfunctions, and appropriate repairs, shall be submitted within 90 days of completing the inspection and no later than **December 12, 2014**. The Discharger shall submit to the Regional Water Board Executive Officer for approval an Outfall Inspection work plan no later than **December 1, 2011**.

## **X. REPORTING REQUIREMENTS**

### **A. General Monitoring and Reporting Requirements**

1. The Discharger shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.
2. Schedules of Compliance. If applicable, the Discharger shall submit all reports and documentation required by compliance schedules that are established by this Order. Such reports and documentation shall be submitted to the Regional Water Board on or before each compliance date established by this Order. If noncompliance is reported, the Discharger shall describe the reasons for noncompliance and a specific date when compliance will be achieved. The Discharger shall notify the Regional Water Board when it returns to compliance with applicable compliance dates established by schedules of compliance.

### **B. Self Monitoring Reports (SMRs)**

1. At any time during the term of this permit, the State or Regional Water Board may notify the Discharger to electronically submit Self-Monitoring Reports (SMRs) using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site (<http://www.waterboards.ca.gov/ciwqs/index.html>). Until such notification is given, the Discharger shall submit hard copy SMRs. The CIWQS Web site will provide additional directions for SMR submittal in the event there will be service interruption for electronic submittal.
2. The Discharger shall report in the SMR the results for all monitoring specified in this MRP under sections III through IX. The Discharger shall submit monthly SMRs including the results of all required monitoring using USEPA-approved test methods or other test methods specified in this Order. If the Discharger monitors any pollutant more frequently than required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the SMR.
3. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

**Table E-8. Monitoring Periods and Reporting Schedule**

| Sampling Frequency   | Monitoring Period Begins On... | Monitoring Period                                                                                                     | SMR Due Date                                                   |
|----------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Continuous           | July 1, 2010                   | All                                                                                                                   | First day of second calendar month following month of sampling |
| Daily                | July 1, 2010                   | Midnight through 11:59 PM or any 24-hour period that reasonably represents a calendar day for purposes of sampling.   | First day of second calendar month following month of sampling |
| Weekly               | July 4, 2010                   | Sunday through Saturday                                                                                               | First day of second calendar month following month of sampling |
| Monthly              | July 1, 2010                   | 1 <sup>st</sup> day of calendar month through last day of calendar month                                              | First day of second calendar month following month of sampling |
| Quarterly            | July 1, 2010                   | January 1 through March 31<br>April 1 through June 30<br>July 1 through September 30<br>October 1 through December 31 | First day of second calendar month following month of sampling |
| Annually             | July 1, 2010                   | January 1 through December 31                                                                                         | March 1 each year                                              |
| Once per Permit Term | January 1, 2011                | January 1 through December 31                                                                                         | January 2, 2015                                                |

4. Reporting Protocols. The Discharger shall report with each sample result the applicable reported Minimum Level (ML) and the current Method Detection Limit (MDL), as determined by the procedure in section 136.

The Discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- a. Sample results greater than or equal to the reported ML shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- b. Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.

For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ as well as the words "Estimated Concentration" (may be shortened to "Est. Conc."). The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy (+ a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.

- c. Sample results less than the laboratory's MDL shall be reported as "Not Detected," or ND.
  - d. Dischargers are to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.
5. The Discharger shall submit SMRs in accordance with the following requirements:
- a. The Discharger shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations. The reported data shall include calculation of all effluent limitations that require averaging, taking of a median or other computation. The Discharger is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format within the system, the Discharger shall electronically submit the data in a tabular format as an attachment.
  - b. The Discharger shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify:
    - (1) Facility name;
    - (2) WDID;
    - (3) Applicable period of monitoring and reporting;
    - (4) Violations of the WDRs (identified violations must include a description of the requirement that was violated and a description of the violation);
    - (5) Corrective actions taken or planned; and
    - (6) The proposed time schedule for corrective actions.
  - c. SMRs must be submitted to the Regional Water Board, signed and certified as required by the Standard Provisions (Attachment D), to the address listed below:

**North Coast Regional Water Board  
5550 Skylane Blvd., Suite A  
Santa Rosa, CA 95403**

### **C. Discharge Monitoring Reports (DMRs)**

- 1. As described in Section X.B.1 above, at any time during the term of this permit, the State or Regional Water Board may notify the Discharger to electronically submit SMRs that will satisfy federal requirements for submittal of Discharge Monitoring

Reports (DMRs). Until such notification is given, the Discharger shall submit DMRs in accordance with the requirements described below

2. DMRs must be signed and certified as required by the standard provisions (Attachment D). The Discharger shall submit the original DMR and one copy of the DMR to the address listed below:

| <b>STANDARD MAIL</b>                                                                                                                     | <b>FEDEX/UPS/<br/>OTHER PRIVATE CARRIERS</b>                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| State Water Resources Control Board<br>Division of Water Quality<br>c/o DMR Processing Center<br>PO Box 100<br>Sacramento, CA 95812-1000 | State Water Resources Control Board<br>Division of Water Quality<br>c/o DMR Processing Center<br>1001 I Street, 15 <sup>th</sup> Floor<br>Sacramento, CA 95814 |

3. All discharge monitoring results must be reported on the official USEPA pre-printed DMR forms (EPA Form 3320-1). Forms that are self-generated will not be accepted unless they follow the exact same format of EPA Form 3320-1.

#### **D. Other Reports**

1. The Discharger shall report the results of any special studies, acute and chronic toxicity testing, TRE/TIE, PMP, and Pollution Prevention Plan required by Special Provisions – VI.C.2 and 3 of this Order. The Discharger shall submit reports with the first monthly SMR scheduled to be submitted on or immediately following the report due date in compliance with SMR reporting requirements described in subsection X.B. above.
2. Annual Report. The Discharger shall submit an Annual Report to the Regional Water Board for each calendar year. The report shall be submitted by March 1st of the following year. The report shall, at a minimum, include the following:
  - a. Both tabular and, where appropriate, graphical summaries of the monitoring data and disposal records from the previous year. If the Discharger monitors any pollutant more frequently than required by this Order, using test procedures approved under Part 136 or as specified in this Order, the results of this monitoring shall be included in the calculation and a report of the data submitted with the SMR.
  - b. A comprehensive discussion of the facility's compliance (or lack thereof) with all effluent limitations and other WDRs, and the corrective actions taken or planned, which may be needed to bring the discharge into full compliance with the Order.
  - c. A summary report of source control activities completed during the calendar year, in accordance with Special Provisions, VI.C.5.b of the Order.

- d. A summary report, if applicable, of the amount of sludge or biosolids placed in a landfill and the landfill(s) which received the sludge or biosolids, in accordance with Special Provisions, VI.C.5.c(4) of the Order.

## E. Spills and Overflows Notification

1. All spills and sanitary sewer overflows (SSOs) equal to or in excess of 1,000 gallons or any size spill or SSO that results in a discharge to a drainage channel or a surface water:
  - a. As soon as possible, but not later than **two (2) hours** after becoming aware of the discharge, the Discharger shall notify the State Office of Emergency Services (OES), the local health officer or directors of environmental health with jurisdiction over affected water bodies or land areas, and the Regional Water Board<sup>7</sup>.

Information to be provided verbally to the Regional Water Board includes:

- i. Name and contact information of caller;
    - ii. Date, time and location of spill occurrence;
    - iii. Estimates of spill volume, rate of flow, and spill duration;
    - iv. Surface water bodies impacted, if any;
    - v. Cause of spill;
    - vi. Cleanup actions taken or repairs made; and
    - vii. Responding agencies.
  - b. As soon as possible, but not later than **twenty-four (24) hours** after becoming aware of a discharge, the Discharger shall submit to the Regional Water Board a certification that the State Office of Emergency Services and the local health officer or directors of environmental health with jurisdiction over affected water bodies or land areas have been notified of the discharge. For the purpose of this requirement, "certification" means an OES certification number and, for the local health department, name of local health staff, department name, phone number and date and time contacted.
  - c. Within **five (5) business days**, the Discharger shall submit a written report to the Regional Water Board office. The report must include all available details related to the cause of the spill and corrective action taken or planned to be taken, as well as copies of reports submitted to other agencies.

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<sup>7</sup> The contact number for spill reporting for the Office of Emergency Services is (800) 852-7550. The contact number of the Regional Water Board during normal business hours is (707) 576-2220. After normal business hours, spill reporting to OES will satisfy the 2 hour notification requirement for the Regional Water Board.

Information to be provided in writing includes:

- i. Information provided in verbal notification;
    - ii. Other agencies notified by phone;
    - iii. Detailed description of cleanup actions and repairs taken; and
    - iv. Description of actions that will be taken to minimize or prevent future spills.
  - d. In the cover letter of the monthly monitoring report, the Discharger shall include a brief written summary of the event and any additional details related to the cause or resolution of the event, including, but not limited to results of any water quality monitoring conducted.
2. Discharges less than 1,000 gallons that do not reach a drainage channel or a surface water:
- a. As soon as possible, but not later than **twenty-four (24) hours** after becoming aware of the discharge, the Discharger shall notify the Regional Water Board and provide the applicable information specified in requirement 1.A of this section.
  - b. In the cover letter of the monthly monitoring report, the Discharger shall include a written description of the event.

## ATTACHMENT F – FACT SHEET

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## ATTACHMENT F – FACT SHEET

As described in section II of this Order, this Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for dischargers in California. Only those sections or subsections of this Order that are specifically identified as “not applicable” have been determined not to apply to this Discharger. Sections or subsections of this Order not specifically identified as “not applicable” are fully applicable to this Discharger.

### I. PERMIT INFORMATION

The following table summarizes administrative information related to the facility.

**Table F-1. Facility Information**

|                                                     |                                                              |
|-----------------------------------------------------|--------------------------------------------------------------|
| <b>WDID</b>                                         | 1B83129OMEN                                                  |
| <b>Discharger</b>                                   | Mendocino City Community Services District (CSD)             |
| <b>Name of Facility</b>                             | Mendocino CSD Wastewater Treatment Facility (WWTF)           |
| <b>Facility Address</b>                             | 10500 Kelly Street                                           |
|                                                     | Mendocino, CA 95460                                          |
|                                                     | Mendocino County                                             |
| <b>Facility Contact, Title and Phone</b>            | Michael Kelley, District Superintendent, (707) 937-5790      |
| <b>Authorized Person to Sign and Submit Reports</b> | Same as above                                                |
| <b>Mailing Address</b>                              | P.O. Box 1029, Mendocino CA 95460                            |
| <b>Billing Address</b>                              | Same as above                                                |
| <b>Type of Facility</b>                             | Publicly Owned Treatment Works (POTW)                        |
| <b>Major or Minor Facility</b>                      | Minor                                                        |
| <b>Threat to Water Quality</b>                      | 2                                                            |
| <b>Complexity</b>                                   | B                                                            |
| <b>Pretreatment Program</b>                         | No                                                           |
| <b>Reclamation Requirements</b>                     | Yes                                                          |
| <b>Facility Permitted Flow</b>                      | 0.3 million gallons per day (mgd) (average dry weather flow) |
| <b>Facility Design Flow</b>                         | 0.3 mgd (average dry weather treatment capacity)             |
| <b>Watershed</b>                                    | Mendocino Coast                                              |
| <b>Receiving Water</b>                              | Pacific Ocean                                                |
| <b>Receiving Water Type</b>                         | Marine                                                       |

- A. Mendocino City CSD (hereinafter Discharger) owns and operates the Mendocino City CSD WWTF, a POTW.

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and State laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

- B. The treatment facility discharges treated wastewater to the Pacific Ocean, waters of the United States, and is currently regulated by Regional Water Board Order No. R1-2004-0055, which was adopted on August 25, 2004 and expired on August 25, 2009. The terms and conditions of the current Order have been automatically continued and remain in effect until new waste discharge requirements and National Pollutant Discharge Elimination System (NPDES) permit are adopted pursuant to this Order.
- C. The Discharger filed a Report of Waste Discharge (ROWD) and submitted an application dated September 8, 2008 for renewal of its Waste Discharge Requirements (WDRs) and National Pollutant Discharge Elimination System (NPDES) permit.

## **II. FACILITY DESCRIPTION**

Mendocino City CSD owns and operates the wastewater collection, treatment, and disposal facilities that serve approximately 4,000 people, including 1,000 full-time residents and many visitors and tourists to the City, Russian Gulch State Park, and Headlands State Park. The WWTF is located at 10500 Kelly Street in Mendocino, Mendocino County, California.

### **A. Description of Wastewater Treatment or Controls**

The WWTF treats domestic and commercial wastewater. The treatment facility has an average dry weather treatment capacity of 0.3 mgd.

The wastewater collection system includes approximately 12.5 miles of vitrified clay and PVC pipe, ranging in size from 15 to 4 inches, and three lift stations.

Wastewater treatment processes include comminution, extended aeration activated sludge, secondary clarification, and tertiary filtration. Effluent is chlorinated, dechlorinated, and flows by gravity to a flow equalization pond. Influent flow in excess of the design flow can be routed to a 300,000 gallon overflow pond for storage until flows diminish, when the excess flow is routed back to the headworks for treatment. Effluent from the equalization pond is controlled at flow control structure A by sliding stop gates, from which flow continues to flow control structure B, which has two pressure control valves to prevent backflow conditions in the outfall. From flow control structure B, effluent is discharged to the Pacific Ocean at Discharge Point 001 via a 996 foot outfall designed to achieve a minimal dilution of 100:1. Onsite storm water runoff is diverted to the plant headworks for treatment.

Solids are aerobically digested, dewatered to approximately 12 percent solids by a belt press, and dried to approximately 90 percent solids using a sludge dryer. Condensate from the dryer is directed to the plant headworks for treatment. Biosolids are placed in a landfill for final disposal.

During the dry weather season (generally May through October) and other periods, as needed, tertiary treated effluent is discharged to the water recycling system (Discharge Point 002), which consists of a 55,000 gallon storage tank, accompanying appurtenances, and a pop-up sprinkler system located on the Mendocino High School athletic fields. The recycled water system is regulated under Regional Water Board Order No. 97-66.

## **B. Discharge Points and Receiving Waters**

The treatment facility's point of discharge to the Pacific Ocean at Discharge Point 001 is located within the Mendocino Coast Hydrologic Unit at 39° 18' 21" N latitude and 123° 48' 30" W longitude. The outfall structure is an 8-inch diameter pipe, 996 feet in length, which provides minimum initial dilution of 100 to 1.

## **C. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data**

Effluent limitations contained in the existing Order for discharges from Discharge Point 001 (Monitoring Location EFF-001) for conventional, non-conventional, and toxic pollutants and representative monitoring data from the term of the previous Order are summarized as follows.

**Table F-2. Historic Effluent Limitations and Monitoring Data – Discharge Point 001**

| Parameter               | Units      | Effluent Limitation    |                |                  |                |                       | Monitoring Data<br>(1/2005 - 12/2008) |
|-------------------------|------------|------------------------|----------------|------------------|----------------|-----------------------|---------------------------------------|
|                         |            | Monthly Average        | Weekly Average | Daily Maximum    | 6-Month Median | Instantaneous Maximum | Highest Average Monthly <sup>1</sup>  |
| BOD <sub>5</sub>        | mg/L       | 30                     | 45             | ---              | ---            | ---                   | 5.6                                   |
| Percent Removal, BOD    | %          | ≥85                    | ---            | ---              | ---            | ---                   | Minimum - 99.0                        |
| TSS                     | mg/L       | 30                     | 45             | ---              | ---            | ---                   | 17.1                                  |
| Percent Removal, TSS    | %          | ≥85                    | ---            | ---              | ---            | ---                   | Minimum - 97.6                        |
| pH                      | s.u.       | 6.0 – 9.0 at all times |                |                  |                |                       | Minimum - 5.9<br>Maximum - 7.1        |
| Settleable Solids       | mL/L-hr    | 1.0                    | 1.5            | 3.0              | ---            | ---                   | <0.1                                  |
| Oil and Grease          | mg/L       | 25                     | 40             | 75               | ---            | ---                   | 15                                    |
| Turbidity               | NTU        | 75                     | 100            | 225              | ---            | ---                   | 4.1                                   |
| Total Coliform Bacteria | MPN/100 mL | 70                     | ---            | 230 <sup>2</sup> | ---            | ---                   | 1600                                  |
| Copper                  | µg/L       | ---                    | ---            | 1012             | 103            | 2830                  | 73<br>(Daily Max)                     |
|                         | lbs/day    | ---                    | ---            | 2.5              | 0.26           | 7.1                   | ---                                   |
| TCDD Equivalents        | µg/L       | 4E-07                  | ---            | ---              | ---            | ---                   | 8.7E-07<br>(Daily Max)                |
|                         | lbs/day    | 1E-10                  | ---            | ---              | ---            | ---                   | ---                                   |

#### D. Compliance Summary

The Discharger has been subject to one Notice of Violation and one Administrative Civil Liability (ACL) Complaint during the term of the previous permit. NOV Order No. R1-2008-0084 listed a pH limitation violation and a daily coliform limitation violation of limitations contained in Order No. R1-2004-0055; however, these violations were considered infrequent and not serious, and no Mandatory Minimum Penalties (MMPs) were assessed.

<sup>1</sup> Unless otherwise noted.

<sup>2</sup> Not more than 10 percent of samples shall exceed 230 MPN/100 mL.

The Discharger has been in compliance with all other effluent limitations, except the effluent limitation for TCDD equivalents, during the term of the previous Order. Since Order No. R1-2004-0055 was adopted, the discharger has had one exceedance of the effluent limitation for TCDD equivalents. ACL Complaint No. R1-2009-0018 assessed civil liability for a serious violation of the effluent limitation for TCDD Equivalents contained in Order No. R1-2004-0055 and for a violation of the prohibition regarding the discharge of untreated or partially treated wastewater that occurred as a sanitary sewer overflow (SSO) in 2008. An MMP of \$3,000 was assessed.

#### **E. Planned Changes**

There are no planned changes at the Mendocino City CSD WWTF.

### **III. APPLICABLE PLANS, POLICIES, AND REGULATIONS**

The requirements contained in the proposed Order are based on the requirements and authorities described in this section. This section provides supplemental information, where appropriate, for the plans, policies, and regulations relevant to the discharge.

#### **A. Legal Authorities**

This Order is issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. Environmental Protection Agency (USEPA) and chapter 5.5, division 7 of the California Water Code (commencing with section 13370). It shall serve as a NPDES permit for point source discharges from this facility to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to article 4, chapter 4, division 7 of the Water Code (commencing with section 13260).

#### **B. California Environmental Quality Act (CEQA)**

Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of CEQA, Public Resources Code sections 21100 through 21177.

#### **C. State and Federal Regulations, Policies, and Plans**

- 1. Water Quality Control Plans.** The Regional Water Quality Control Board (Regional Water Board) adopted a *Water Quality Control Plan for the North Coast Region* (hereinafter Basin Plan) that designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. In addition, the Basin Plan implements State Water Resources Control Board (State Water Board) Resolution No. 88-63, which establishes State policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply. Total dissolved solids concentrations in ocean waters exceed 3,000 mg/L, and thereby meet an exception to Resolution 88-63. The “municipal or domestic supply” (MUN) designation is therefore not applicable to the coastal receiving water

for discharges at Discharge Point 001. Beneficial uses applicable to the receiving water for discharges from the Mendocino City CSD WWTF are as follows.

**Table F-3. Basin Plan Beneficial Uses**

| Discharge Point | Receiving Water | Beneficial Uses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 001             | Pacific Ocean   | <p><b>Existing:</b></p> <ul style="list-style-type: none"> <li>• Navigation (NAV)</li> <li>• Water Contact Recreation (REC1)</li> <li>• Non-Contact Water Recreation (REC2)</li> <li>• Commercial and Sport Fishing (COMM)</li> <li>• Marine Habitat (MAR)</li> <li>• Wildlife Habitat (WILD)</li> <li>• Preservation of Rare, Threatened, or Endangered Species (RARE)</li> <li>• Migration of Aquatic Organisms (MIGR)</li> <li>• Spawning, Reproduction, and/or Early Development (SPWN)</li> <li>• Shellfish Harvesting (SHELL)</li> <li>• Aquaculture (AQUA)</li> </ul> <p><b>Potential:</b></p> <ul style="list-style-type: none"> <li>• Industrial Service Supply (IND)</li> <li>• Industrial Process Supply (PRO)</li> <li>• Preservation of Areas of Special Biological Significance (ASBS)</li> </ul> |

Requirements of this Order implement the Basin Plan.

- 2. Thermal Plan.** The State Water Board adopted a *Water Quality Control Plan for Control of Temperature in the Coastal and Interstate Water and Enclosed Bays and Estuaries of California* (Thermal Plan) on May 18, 1972, and amended this plan on September 18, 1975. Requirements of the Order implement the Thermal Plan.
- 3. California Ocean Plan.** The State Water Board adopted the *Water Quality Control Plan for Ocean Waters of California, California Ocean Plan* (Ocean Plan) in 1972 and amended it in 1978, 1983, 1988, 1990, 1997, 2000, and 2005. The State Water Board adopted the latest amendment on April 21, 2005 and it became effective on February 14, 2006. The Ocean Plan is applicable, in its entirety, to point source discharges to the ocean. The Ocean Plan identifies beneficial uses of ocean waters of the State to be protected as summarized below.

**Table F-4. Ocean Plan Beneficial Uses**

| Discharge Point | Receiving Water | Beneficial Uses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 001             | Pacific Ocean   | <ul style="list-style-type: none"> <li>• Industrial Water Supply;</li> <li>• Water Contact and Non-Contact Recreation, Including Aesthetic Enjoyment;</li> <li>• Navigation;</li> <li>• Commercial and Sport Fishing;</li> <li>• Mariculture;</li> <li>• Preservation and Enhancement of Designated Areas of Special Biological Significance (ASBS);</li> <li>• Rare and Endangered Species;</li> <li>• Marine Habitat;</li> <li>• Fish Migration</li> <li>• Fish Spawning; and</li> <li>• Shellfish Harvesting</li> </ul> |

4. **Alaska Rule.** On March 30, 2000, USEPA revised its regulation that specifies when new and revised state and tribal water quality standards (WQS) become effective for CWA purposes [40 C.F.R. § 131.21, 65 Fed. Reg. 24641 (April 27, 2000)]. Under the revised regulation (also known as the Alaska Rule), new and revised standards submitted to USEPA after May 30, 2000, must be approved by USEPA before being used for CWA purposes. The final rule also provides that standards already in effect and submitted to USEPA by May 30, 2000, may be used for CWA purposes, whether or not approved by USEPA.
5. **Antidegradation Policy.** Section 131.12 requires that state water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution No. 68-16. Resolution No. 68-16 incorporates the federal antidegradation policy where the federal policy applies under federal law. Resolution No. 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Regional Water Board's Basin Plan implements, and incorporates by reference, both the State and federal antidegradation policies. The permitted discharge must be consistent with the antidegradation provision of section 131.12 and State Water Board Resolution No. 68-16.
6. **Anti-Backsliding Requirements.** Sections 402(o)(2) and 303(d)(4) of the CWA and federal regulations at title 40, Code of Federal Regulations<sup>3</sup> section 122.44(l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require

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<sup>3</sup> All further statutory references are to title 40 of the Code of Federal Regulations unless otherwise indicated.

that effluent limitations in a reissued permit must be as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed.

#### **D. Impaired Water Bodies on CWA 303(d) List**

Section 303(d) of the federal CWA requires states to identify waterbodies that do not meet water quality standards and are not supporting their beneficial uses after implementation of technology-based effluent limitations on point sources. Each state must submit an updated list, the 303(d) List of Impaired Waterbodies, to USEPA by April of each even numbered year. In addition to identifying the waterbodies that are not supporting beneficial uses, the 303(d) list also identifies the pollutant or stressor causing impairment and establishes a schedule for developing a control plan to address the impairment. The USEPA requires the Regional Water Board to develop total maximum daily loads (TMDLs) for each 303(d) listed pollutant and water body contaminant. TMDLs establish the maximum quantity of a given pollutant that can be added to a water body from all sources without exceeding the applicable water quality standard for that pollutant and determine wasteload allocations (the portion of a TMDL allocated to existing and future point sources) for point sources and load allocations (the portion of a TMDL attributed to existing and future nonpoint sources) for nonpoint sources.

In June 2007, the USEPA provided final approval of the 303(d) list of impaired water bodies prepared by the State. The coastal waters which are the receiving waters for this discharge are not listed on the 303(d) list as being impaired.

#### **E. Other Plans, Policies and Regulations**

1. On April 17, 1997, the State Water Board adopted State Water Board Water Quality Order No. 97-03-DWQ, NPDES General Permit No. CAS000001, Waste Discharge Requirements for Discharges of Storm Water Associated with Industrial Activities Excluding Construction Activities. The general permit is not applicable to the Mendocino CSD.
2. On May 2, 2006, the State Water Board adopted State Water Board Order 2006-0003-DWQ, Statewide General WDRs for Sanitary Sewer Systems. Order No. 2006-0003-DWQ requires that all public agencies that currently own or operate sanitary sewer systems greater than one mile in length that collect and convey untreated or partially treated wastewater to a POTW apply for coverage under the General WDRs. The Discharger has applied for coverage and is subject to the requirements of Order No. 2006-0003-DWQ and any future revisions thereto for operation of its wastewater collection system.
3. On July 22, 2004, the State Water Board adopted State Water Board Order No. 2004-0012-DWQ, General Waste Discharge Requirements for the Discharge of Biosolids to Land for Use as a Soil Amendment in Agricultural, Silvicultural, Horticultural, and Land Reclamation Activities. Should the Discharger choose to

discharge biosolids to land for use as a soil amendment, the Order requires the Discharger to obtain coverage under Order No. 2004-0012-DWQ or other WDRs issued by the Regional Water Board prior to land application.

#### **IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS**

The CWA requires point source dischargers to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. There are two principal bases for effluent limitations in the Code of Federal Regulations: section 122.44(a) requires that permits include applicable technology-based limitations and standards; and section 122.44(d) requires that permits include water quality-based effluent limitations to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water.

##### **A. Discharge Prohibitions**

1. **Discharge Prohibition III.A.** The discharge of any waste not disclosed by the Discharger or not within the reasonable contemplation of the Regional Water Board is prohibited.

This prohibition is retained from the previous Order (Order No. R1-2004-0055) and is based on the Basin Plan, and State Water Board Order WQO 2002-0012 regarding the petition of WDRs Order No. 01-072 for the East Bay Municipal Utility District and Bay Area Clean Water Agencies. In State Water Board Order No. WQO 2002-0012, the State Water Board found that this prohibition is acceptable in Orders, but should be interpreted to apply only to constituents that are either not disclosed by the Discharger, or are not reasonably anticipated to be present in the discharge but have not been disclosed by the Discharger. It specifically does not apply to constituents in the discharge that do not have “reasonable potential” to exceed water quality objectives.

The State Water Board has stated that the only pollutants not covered by this prohibition are those which were “disclosed to the permitting authority and ... can be reasonably contemplated.” [In re the Petition of East Bay Municipal Utilities District et al., (State Water Board, 2002) Order No. WQO 2002-0012, p. 24] In that Order, the State Water Board cited a case which held the Discharger is liable for discharge of pollutants not “within the reasonable contemplation of the permitting authority..., whether spills or otherwise,...” [Piney Run Preservation Assn. v. County Commissioners of Carroll County, Maryland (4<sup>th</sup> Cir. 2001) 268 F. 3d 255, 268.] Thus the State Water Board authority provides that, to be permissible, the constituent discharged (1) must have been disclosed by the Discharger and (2) can be reasonably contemplated by the Regional Water Board.

Whether or not the Discharger reasonably contemplates the discharge of a constituent is not relevant. What matters is whether the Discharger disclosed the constituent to the Regional Water Board or whether the presence of the pollutant in the discharge can otherwise be reasonably contemplated by the Regional Water Board at the time of Order adoption.

2. **Discharge Prohibition III.B.** Creation of pollution, contamination, or nuisance, as defined by Section 13050 of the California Water Code is prohibited.

This prohibition is retained from the previous Order (Order No. R1-2004-0055) and is based on section 13050 of the Water Code.

3. **Discharge Prohibition III.C.** The discharge of sludge or digester supernatant is prohibited, except as authorized under section VI.C.5.c of the Order. (Sludge Disposal and Handling Requirements)

This prohibition is retained from the previous Order (Order No. R1-2004-0055) and is based in restrictions on the disposal of sewage sludge found in federal regulations [40 CFR Part 503 (Biosolids), Part 527 and Part 258] and Title 27 of the California Code of Regulations (CCR).

4. **Discharge Prohibition III.D.** The discharge of untreated or partially treated waste (receiving a lower level of treatment than described in section II.B of the Order) from anywhere within the collection, treatment, or disposal systems is prohibited, except as provided for in Prohibition III. E. and in Attachment D, Standard Provisions (Bypass).

This prohibition has been retained from the previous Order (Order No. R1-2004-0055) and is based on the need to protect beneficial uses of the receiving water from unpermitted discharges, and the intent of the Water Code sections 13260 through 13264 relating to the discharge of waste to waters of the State without filing for and being issued an Order. This prohibition applies to spills not related to sanitary sewer overflows (SSOs) and other unauthorized discharges of wastewater within the collection, treatment, and disposal facilities. The discharge of untreated or partially treated wastewater from the collection, treatment, or disposal facility represents an unauthorized bypass pursuant to section 122.41(m) or an unauthorized discharge which poses a threat to human health and/or aquatic life, and is therefore explicitly prohibited by the Order.

5. **Discharge Prohibition III.E.** Any SSO that results in a discharge of untreated or partially treated wastewater to (a) waters of the State, (b) groundwater, or (c) land that creates pollution, contamination, or nuisance, as defined in Water Code section 13050(m) is prohibited.

This prohibition is established by this Order. The prohibition applies to spills related to SSOs and is based on State standards, including section 13050 of the Water Code and the Basin Plan. This prohibition is consistent with the States' antidegradation policy as specified in State Water Board Resolution No. 68-16 (*Statement of Policy with Respect to Maintaining High Quality of Water in California*) in that the prohibition imposes conditions to prevent impacts to water quality, the degradation of water quality, negative effects on receiving water beneficial uses, and lessening of water quality beyond that prescribed in State Water Board or Regional Water Board plans and policies.

This prohibition is stricter than the prohibitions stated in State Water Board Order 2006-003-DWQ, Statewide General Waste Discharge Requirements for Sanitary Sewer Systems. Order No. 2006-0003-DWQ prohibits SSOs that result in the discharge of untreated or partially treated wastewater to waters of the United States and SSOs that cause a nuisance, compared to Prohibition III.E. of this Order, which prohibits SSO discharges that create nuisance or pollution to waters of the State, groundwater, and land, and which will provide a more complete protection of human health. The rationale for this more strict prohibition is because of the prevalence of high groundwater in the North Coast Region, and this Region's reliance on groundwater as a drinking water source.

6. **Discharge Prohibition III.F.** Discharge of more than 0.3 mgd as an average dry weather flow, as determined from the lowest average daily flow measured over 30 consecutive days, is prohibited.

This prohibition is retained from the previous permit. The facility average dry weather design capacity is 0.3 mgd. Exceedance of the treatment plant's average dry weather flow design capacity may result in lowering the reliability of achieving compliance with water quality requirements.

7. **Discharge Prohibition III.G.** The discharge of recycled, filtered wastewater to a use area other than one designated for that purpose is prohibited.

This prohibition is retained from the previous permit. A discharge of recycled wastewater other than that described in the Order as Discharge Point 002 is an unauthorized discharge and is therefore prohibited.

8. **Discharge Prohibition III.H.** Discharge of any radiological, chemical, or biological warfare agent, or high-level radioactive waste into the ocean is prohibited.

This prohibition is established by this Order and is based on the discharge prohibitions contained in the Ocean Plan.

9. **Discharge Prohibition III.I.** Discharge of sludge directly into the ocean or into a waste stream that discharges to the ocean is prohibited.

This prohibition is newly established by the Order and is based on the discharge prohibitions contained in the Ocean Plan.

## B. Technology-Based Effluent Limitations

### 1. Scope and Authority

Section 301(b) of the CWA and implementing USEPA permit regulations at section 122.44, require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge authorized by this Order must meet minimum federal technology-based requirements based on Secondary Treatment Standards at Part 133 and/ or Best Professional Judgment (BPJ) in accordance with Part 125, section 125.3.

At section 133.102 the Secondary Treatment Standards establish the following minimum level of effluent quality attainable by secondary treatment, which the Regional Water Board must include as minimum effluent limitations in permits issued to POTWs.

**Table F-5. Secondary Treatment Standards from the Federal Regulations**

| Parameter                | Effluent Quality                                                                                   |               |
|--------------------------|----------------------------------------------------------------------------------------------------|---------------|
|                          | 30 Day Average                                                                                     | 7 Day Average |
| BOD <sub>5</sub>         | 30 mg/L                                                                                            | 45 mg/L       |
| TSS                      | 30 mg/L                                                                                            | 45 mg/L       |
| BOD <sub>5</sub> and TSS | The 30 day average percent removal for BOD <sub>5</sub> and TSS shall not be less than 85 percent. |               |
| pH                       | 6.0 – 9.0 at all times                                                                             |               |

In addition, the State Water Board, in Table A of the Ocean Plan, has established technology-based requirements, applicable to all POTWs for suspended solids, settleable solids, turbidity, pH, and grease and oil.

### 2. Applicable Technology-Based Effluent Limitations

Technology-based limitations established by the Order for Discharge Point 001 are summarized in the following table.

**Table F-6. Effluent Limitations for Discharge Point 001**

| Parameter                | Units                | Effluent Limitations   |                |               |
|--------------------------|----------------------|------------------------|----------------|---------------|
|                          |                      | Average Monthly        | Average Weekly | Maximum Daily |
| BOD <sub>5</sub>         | mg/L                 | 30                     | 45             | ---           |
|                          | lbs/day <sup>4</sup> | 75                     | 113            | ---           |
| TSS                      | mg/L                 | 30                     | 45             | ---           |
|                          | lbs/day <sup>3</sup> | 75                     | 113            | ---           |
| BOD <sub>5</sub> and TSS | % Removal            | 85                     | ---            | ---           |
| pH                       | s.u.                 | 6.0 – 9.0 at all times |                |               |
| Settleable Solids        | mL/L-hr              | 1.0                    | 1.5            | 3.0           |
| Grease and Oil           | mg/L                 | 25                     | 40             | 75            |
| Turbidity                | NTU                  | 75                     | 100            | 225           |

a. Discharge Point 001.

- (1) Numeric effluent limitations for BOD<sub>5</sub>, TSS, including the percent removal requirement, and for pH are retained from the previous Order and reflect secondary treatment standards at Part 133. Mass-based limits are required pursuant to section 122.45(f) for the purpose of assuring that dilution is not used as a method of achieving the concentration limitations in the permit. The mass-based limitations were determined using the average dry weather design capacity of the facility. Review of the facility's flow data from 1979-2007 shows that discharge rates in all seasons have not exceeded this design capacity in that time period, and therefore the Regional Water Board has determined that year round compliance with these mass-based limitations is achievable. The Order does not include mass-based limitations for pH because limitations for pH cannot appropriately be expressed in terms of mass.
- (2) Numeric effluent limitations for turbidity are retained from the previous Order and are based in Table A of the Ocean Plan. Ocean Plan Table A effluent limitations for turbidity reflect the minimum level of treatment acceptable under the Ocean Plan, and define reasonable treatment and waste control technology. The Order does not include mass-based limitations for turbidity because limitations for turbidity cannot appropriately be expressed in terms of mass.

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<sup>4</sup> Dry weather mass-based effluent limitations are based on the average dry weather design flow of 0.3 mgd.

- (3) Numeric effluent limitations for oil and grease are retained from the previous Order and reflect the minimum level of treatment acceptable under the Ocean Plan, and define reasonable treatment and waste control technology. This Order does not include mass-based limits for grease and oil because the concentrations-based limitations included in the Order are adequate to ensure compliance with Ocean Plan Table A limitations.
- (4) Effluent limitations for settleable solids have been retained from the previous Order. Settleable solids generally constitute 40 to 65 percent of the suspended solids in domestic wastewaters and are measured volumetrically by quiescent settling of a one liter sample for one hour in an Imhoff cone (and are therefore expressed as mL/L/hr). Method SM 2540F for the analysis of settleable solids describes a lower limit of measurement of settleable solids at 0.1 mL/L/hr, and therefore, the monthly average limitation established by this Order, reflects, in effect, a non-detectable (100 percent removal efficiency) level of settleable solids in the discharge. The Regional Water Board has determined that secondary treatment should remove settleable solids to non-detect levels, and therefore effluent limitations for this parameter are necessary to ensure efficient operation of the treatment facility in addition to ensuring protection of aquatic life from adverse impacts of settleable material in the discharge. The Order does not include mass-based limitations for settleable solids because limitations for settleable solids cannot appropriately be expressed in terms of mass.

## **C. Water Quality-Based Effluent Limitations (WQBELs)**

### **1. Scope and Authority**

Section 301(b) of the CWA and section 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards. This Order contains requirements more stringent than secondary treatment requirements that are necessary to meet applicable water quality standards.

NPDES regulations at section 122.44(d)(1)(i) mandate that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. A reasonable potential analysis (RPA) demonstrated reasonable potential for discharges from the Mendocino City CSD WWTF to cause or contribute to exceedances of applicable water quality criteria for chloroform, dichlorobromomethane, and TCDD equivalents at Discharge Point 001. In addition, reasonable potential was determined without analysis of effluent data for total

coliform bacteria because the facility is a POTW treating domestic waste, and for total residual chlorine, because the Discharger uses chlorine for effluent disinfection.

Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, water quality-based effluent limitations (WQBELs) must be established using: (1) USEPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in section 122.44(d)(1)(vi).

The process for determining reasonable potential and calculating WQBELs when necessary is intended to protect the designated uses of the receiving water as specified in the Basin Plan and Ocean Plan, and achieve applicable water quality objectives and criteria that are contained in other State plans and policies.

## **2. Applicable Beneficial Uses and Water Quality Criteria and Objectives**

- a. Beneficial Uses. Beneficial use designations established by the Basin Plan and Ocean Plan applicable to coastal receiving waters for discharges from the Mendocino City CSD WWTF are discussed in Findings II. H and I of the Order and sections III.C.1 and 2 of this Fact Sheet.
- b. California Ocean Plan. Water quality objectives applicable to discharges at Discharge Point 001 include the water quality objectives established in the Ocean Plan.

## **3. Determining the Need for WQBELs**

### **a. Non-Table B Ocean Plan Water Quality Objectives**

#### **i. Total Coliform Bacteria**

The Ocean Plan includes bacteriological objectives for ocean waters used for water contact recreation and shellfish harvesting. For total and fecal coliform bacteria and the Enterococcus group of bacteria, water contact standards must be met within a zone bounded by the shoreline and a distance of 1,000 feet from the shoreline or the 30-foot depth contour, whichever is further from the shoreline. Shellfish harvesting standards for total coliform bacteria must be maintained throughout the water column. In lieu of receiving water bacteriological monitoring, the Discharger has been required to meet the most stringent water quality standards (shellfish harvesting standards) at end-of-pipe.

Because shellfish harvesting is a designated beneficial use of the receiving water, and because of the presence of coliform bacteria in domestic wastewater effluent, Regional Water Board staff have therefore determined that there is reasonable potential that the discharge can cause or contribute to exceedances of the Ocean Plan water quality objectives for total coliform bacteria for shellfish harvesting.

Because reasonable potential is demonstrated for total coliform bacteria, the applicable total coliform bacteria water quality objectives from the Ocean Plan are established as end-of-pipe effluent limitations at Discharge Point 001, as described below.

The disinfected effluent at Discharge Point 001, as measured at EFF-002, shall not contain concentrations of total coliform bacteria exceeding the following limitations.

- (1) The median concentration shall not exceed a Most Probable Number (MPN) of 70 per 100 milliliters (mL) for a calendar month.
- (2) Not more than 10 percent of samples shall exceed an MPN of 230 per 100 mL, using bacteriological results from ten consecutive samples. This limitation is to be applied as a moving determination of daily values for any set of ten consecutive samples using simple order statistics. For example, for weekly sampling results ( $N = 10$  weekly samples) where the ordered set is  $\{<2, <2, <2, 2, 4, 4, 13, 23, 180, 230\}$ , the 90<sup>th</sup> percentile value is 180. Since the value of 180 MPN/ 100 mL does not exceed the Ocean Plan shellfish standard of not more than 10 percent of samples exceeding 230 MPN/ 100 mL, the discharger is in compliance.

#### **b. Ocean Plan Table B Water Quality Objectives**

Procedures for performing a Reasonable Potential Analysis (RPA) for toxic material in ocean discharges are described in Section III.C. and Appendix VI of the Ocean Plan. In general, the procedure is a statistical method that projects an effluent data set while taking into account the averaging period of water quality objectives, the long term variability of pollutants in the effluent, limitations associated with sparse data sets, and uncertainty associated with censored data sets. The procedure assumes a lognormal distribution of the effluent data set, and compares the 95<sup>th</sup> percentile concentration at 95 percent confidence of each Table B pollutant, accounting for dilution, to the applicable water quality criterion. The RPA results in one of three following endpoints:

- Endpoint 1 – There is “reasonable potential,” and a WQBEL and monitoring are required.

- Endpoint 2 - There is no “reasonable potential.” WQBELs are not required, and monitoring is required at the discretion of the Regional Water Board.
- Endpoint 3 - The RPA is inconclusive. Existing WQBELs are retained, and monitoring is required.

The State Water Resources Control Board has developed a reasonable potential calculator, which is available at <http://www.waterboards.ca.gov/plnspols/docs/oplans/rpcalc.zip>. The calculator (RPcalc 2.0) was used in conducting the RPA and considers several pathways in the determination of reasonable potential.

#### 1. First Path

If available information about the receiving water or the discharge supports a finding of reasonable potential without analysis of effluent data, the Regional Water Board may decide that WQBELs are necessary after a review of available information. Such information may include: the facility or discharge type, solids loading, lack of dilution, history of compliance problems, potential toxic effects, fish tissue data, 303 (d) status of the receiving water, or the presence of threatened or endangered species or their critical habitat, or other information.

#### 2. Second Path

If any pollutant concentration, adjusted to account for dilution, is greater than the most stringent applicable water quality objective, there is reasonable potential for that pollutant. For Table B pollutants, the adjusted pollutant concentration  $X$  is calculated using the equation  $X = (C_e + D_m C_s) / (D_m + 1)$ , where  $C_e$  is the applicable water quality objective,  $D_m$  is the minimum probable initial dilution expressed as parts seawater per part wastewater and  $C_s$  is the background seawater concentration from Table C of the Ocean Plan.

#### 3. Third Path

If the effluent data contains 3 or more detected and quantified values [i.e., values that are at or above the Minimum Level (ML)], and all values in the data set are at or above the ML, a parametric RPA is conducted to project the range of possible effluent values. The 95<sup>th</sup> percentile concentration is determined at 95 percent confidence for each pollutant, and compared to the most stringent applicable water quality objective to determine reasonable potential. A parametric analysis assumes that the range of possible effluent values is distributed log normally. If the 95<sup>th</sup> percentile

value is greater than the most stringent applicable water quality objective, there is reasonable potential for that pollutant.

#### 4. Fourth Path

If the effluent data contains 3 or more detected and quantified values (i.e., values that are at or above the ML), but at least one value in the data set is less than the ML, a parametric RPA is conducted according to the following steps.

(1) If the number of censored values (those expressed as a “less than” value) account for less than 80 percent of the total number of effluent values, calculate the  $M_L$  (the mean of the natural log of transformed data) and  $S_L$  (the standard deviation of the natural log of transformed data) and conduct a parametric RPA, as described above for the Third Path.

(2) If the number of censored values account for 80 percent or more of the total number of effluent values, conduct a non-parametric RPA, as described below for the Fifth Path. (A non-parametric analysis becomes necessary when the effluent data is limited, and no assumptions can be made regarding its possible distribution.)

#### 5. Fifth Path

A non-parametric RPA is conducted when the effluent data set contains less than 3 detected and quantified values, or when the effluent data set contains 3 or more detected and quantified values but the number of censored values accounts for 80 percent or more of the total number of effluent values. A non-parametric analysis is conducted by ordering the data, comparing each result to the applicable water quality objective, and accounting for ties. The sample number is reduced by one for each tie, when the dilution-adjusted method detection limit (MDL) is greater than the water quality objective. If the adjusted sample number, after accounting for ties, is greater than 15, the pollutant has no reasonable potential to exceed the water quality objective. If the sample number is 15 or less, the RPA is inconclusive, monitoring is required, and any existing effluent limits in the expiring permit are retained.

The following table presents results of the RPA, performed in accordance with procedures described by the Ocean Plan and summarized above, for the Mendocino City CSD WWTF. Here, the RPA was conducted using effluent monitoring data generated during monitoring events in December 2004 through December 2008.

Table F-7 includes the water quality objective for each Table B pollutant, and the maximum observed effluent concentration, adjusted for dilution (here, 100:1). The RPA endpoint for each Table B pollutant is identified. As shown in the following table, the RPA commonly leads to Endpoint 3, meaning that the RPA is inconclusive, when a majority of the effluent data is reported as ND (not detected). In these circumstances, the Regional Water Board views the “inconclusive” result as an indication of no concern for a particular pollutant; however, continued monitoring will be required for those pollutants during the term of the reissued permit.

A parametric RPA found that for chloroform and dichlorobromomethane, the lognormal upper 95 percent confidence bound (UCB) for the 95<sup>th</sup> percentile of the effluent distribution after complete mixing to exceed the water quality objective for these pollutants, resulting in a showing of “reasonable potential” (Endpoint 1); therefore effluent limitations for these pollutants are required for Discharge Point 001. For TCDD equivalents, the maximum observed effluent concentration, adjusted for dilution, exceeded the water quality objective for this pollutant, resulting in a showing of “reasonable potential” (Endpoint 1); therefore, effluent limitations for this pollutant are required for Discharge Point 001. In addition, reasonable potential was determined for total residual chlorine because the Discharger uses chlorine for effluent disinfection.

**Table F-7. RPA Results – Discharge Point 001<sup>5</sup>**

| Table B Pollutant                                       | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Conc. (µg/L) | RPA Result, Comment                                                                                                |
|---------------------------------------------------------|---------------------------|----------------|--------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Objectives for Protection of Marine Aquatic Life</b> |                           |                |                    |                           |                                                                                                                    |
| Ammonia (as N)                                          | 600                       | 48             | 0                  | 98.5                      | Endpoint 2 – An effluent limitation is not required for this pollutant. Monitoring may be required as appropriate. |
| Arsenic                                                 | 8                         | 6              | 5                  | 2.9                       | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Cadmium                                                 | 1                         | 6              | 4                  | 0.006                     | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Chlorinated Phenolics                                   | 1                         | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Chromium (VI)                                           | 2                         | 7              | 6                  | 0.06                      | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |

<sup>5</sup> Notes to Table F-7:

ND indicates that the pollutant was not detected.

Minimum probable initial dilution for this Discharger is 100:1. The Maximum Effluent Concentration is the expected concentration after complete mixing, in accordance with reasonable potential procedure in Appendix VI of the Ocean Plan.

Effluent data used for this RPA are from December 2004 through December 2008.

| Table B Pollutant                                                 | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Conc. (µg/L) | RPA Result, Comment                                                                                                |
|-------------------------------------------------------------------|---------------------------|----------------|--------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| Copper                                                            | 3                         | 50             | 12                 | 2.7                       | Endpoint 2 – An effluent limitation is not required for this pollutant. Monitoring may be required as appropriate. |
| Cyanide                                                           | 1                         | 6              | 6                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Total Chlorine Residual                                           | 202                       | --             | --                 | --                        | Endpoint 1 – Other Information                                                                                     |
| Endosulfan (total)                                                | 0.009                     | 1              | 1                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Endrin                                                            | 0.002                     | 1              | 1                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| HCH                                                               | 0.004                     | 1              | 1                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Lead                                                              | 2                         | 6              | 4                  | 0.02                      | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Mercury                                                           | 0.04                      | 1              | 1                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Nickel                                                            | 5                         | 6              | 6                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Non-chlorinated Phenolics                                         | 30                        | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Selenium                                                          | 15                        | 6              | 6                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Silver                                                            | 0.7                       | 6              | 6                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Zinc                                                              | 20                        | 6              | 0                  | 9.1                       | Endpoint 2 – An effluent limitation is not required for this pollutant. Monitoring may be required as appropriate. |
| <b>Objectives for Protection of Human Health – Noncarcinogens</b> |                           |                |                    |                           |                                                                                                                    |
| 1,1,1-Trichloroethane                                             | 540000                    | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| 2,4-Dinitrophenol                                                 | 4.0                       | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| 2-Methyl-4,6-Dinitrophenol                                        | 220                       | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Acrolein                                                          | 220                       | 1              | 1                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Antimony                                                          | 1200                      | 6              | 6                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Bis(2-Chloroethoxy)Methane                                        | 4.4                       | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Bis(2-Chloroisopropyl)Ether                                       | 1200                      | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Chlorobenzene                                                     | 570                       | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Chromium (III)                                                    | 190000                    | 6              | 6                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Dichlorobenzenes                                                  | 5100                      | 4              | 4                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |
| Diethyl Phthalate                                                 | 33000                     | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                                      |

| <b>Table B Pollutant</b>                                       | <b>Most Stringent WQO (µg/L)</b> | <b>No. of Samples</b> | <b>No. of Non-Detects</b> | <b>Max Effluent Conc. (µg/L)</b> | <b>RPA Result, Comment</b>                                                                               |
|----------------------------------------------------------------|----------------------------------|-----------------------|---------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------|
| Dimethyl Phthalate                                             | 820000                           | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Di-n-Butyl Phthalate                                           | 3500                             | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Ethylbenzene                                                   | 4100                             | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Fluoranthene                                                   | 15                               | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Hexachlorocyclopentadiene                                      | 58                               | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Nitrobenzene                                                   | 4.9                              | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Thallium                                                       | 2                                | 6                     | 6                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Toluene                                                        | 85000                            | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Tributyltin                                                    | 0.0088                           | 1                     | 0                         | 0.00003                          | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| <b>Objectives for Protection of Human Health – Carcinogens</b> |                                  |                       |                           |                                  |                                                                                                          |
| 1,1,2,2-Tetrachloroethane                                      | 2.3                              | 2                     | 2                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| 1,1,2-Trichloroethane                                          | 9.4                              | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| 1,1-Dichloroethylene                                           | 0.9                              | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| 1,2-Dichloroethane                                             | 28                               | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| 1,2-Diphenylhydrazine                                          | 0.16                             | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| 1,3-Dichloropropylene                                          | 8.9                              | 1                     | 1                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| 1,4 Dichlorobenzene                                            | 18                               | 5                     | 5                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| <b>TCDD Equivalents</b>                                        | <b>3.9E-9</b>                    | <b>6</b>              | <b>1</b>                  | <b>8.6E-9</b>                    | <b>Endpoint 1 – An effluent limitation must be developed for this pollutant. Monitoring is required.</b> |
| 2,4,6-Trichlorophenol                                          | 0.29                             | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| 2,4-Dinitrotoluene                                             | 2.6                              | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| 3,3'-Dichlorobenzidine                                         | 0.0081                           | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Acrylonitrile                                                  | 0.10                             | 1                     | 1                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Aldrin                                                         | 2.2E-5                           | 1                     | 1                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Benzene                                                        | 5.9                              | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Benzidine                                                      | 6.9E-5                           | 3                     | 3                         | ND                               | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |

| Table B Pollutant           | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Conc. (µg/L) | RPA Result, Comment                                                                                      |
|-----------------------------|---------------------------|----------------|--------------------|---------------------------|----------------------------------------------------------------------------------------------------------|
| Beryllium                   | 0.033                     | 6              | 6                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Bis(2-Chloroethyl)Ether     | 0.045                     | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Bis(2-Ethylhexyl)Phthalate  | 3.5                       | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Carbon Tetrachloride        | 0.90                      | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Chlordane                   | 2.3E-5                    | 1              | 1                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Chlorodibromomethane        | 8.6                       | 3              | 1                  | 0.04                      | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| <b>Chloroform</b>           | 130                       | <b>3</b>       | <b>0</b>           | <b>0.14</b>               | <b>Endpoint 1 – An effluent limitation must be developed for this pollutant. Monitoring is required.</b> |
| DDT (total)                 | 0.00017                   | 1              | 1                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| <b>Dichlorobromomethane</b> | 6.2                       | <b>3</b>       | <b>0</b>           | <b>0.09</b>               | <b>Endpoint 1 – An effluent limitation must be developed for this pollutant. Monitoring is required.</b> |
| Dieldrin                    | 0.00004                   | 1              | 1                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Halomethanes                | 130                       | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Heptachlor                  | 0.00005                   | 1              | 1                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Heptachlor Epoxide          | 0.00002                   | 1              | 1                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Hexachlorobenzene           | 0.00021                   | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Hexachlorobutadiene         | 14                        | 5              | 5                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Hexachloroethane            | 2.5                       | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Isophorone                  | 730                       | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Methylene Chloride          | 450                       | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| N-Nitrosodimethylamine      | 7.3                       | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| N-Nitrosodi-n-Propylamine   | 0.38                      | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| N-Nitrosodiphenylamine      | 2.5                       | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| PAHs (total)                | 0.0088                    | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| PCBs                        | 1.9E-5                    | 1              | 1                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |
| Tetrachloroethylene         | 2.0                       | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND.                            |

| Table B Pollutant | Most Stringent WQO (µg/L) | No. of Samples | No. of Non-Detects | Max Effluent Conc. (µg/L) | RPA Result, Comment                                                           |
|-------------------|---------------------------|----------------|--------------------|---------------------------|-------------------------------------------------------------------------------|
| Toxaphene         | 0.00021                   | 1              | 1                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND. |
| Trichloroethylene | 27                        | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND. |
| Vinyl Chloride    | 36                        | 3              | 3                  | ND                        | Endpoint 3 – RPA is inconclusive. Less than 3 detects or greater than 80% ND. |

#### 4. WQBEL Calculations

Based on results of the RPA, performed in accordance with methods of the Ocean Plan for discharges to the Pacific Ocean and other available information pertaining to the discharge, the Regional Water Board is establishing WQBELs for chloroform, dichlorobromomethane, TCDD equivalents, and total residual chlorine.

As described by Section III. C of the Ocean Plan, effluent limits for Table B pollutants are calculated according to the following equation.

$$C_e = C_o + D_m (C_o - C_s)$$

Where ...

$C_e$  = the effluent limitation (µg/L)

$C_o$  = the concentration (the water quality objective) to be met at the completion of initial dilution (µg/L).

$C_s$  = background seawater concentration (µg/L)

$D_m$  = minimum probable initial dilution expressed as parts seawater per part wastewater (here,  $D_m$  = 100)

For the Mendocino City CSD WWTF,  $D_m$  (100) is unchanged from the previous Order (No. R1-2004-0055). Initial dilution is the process that results in the rapid and irreversible turbulent mixing of wastewater with ocean water around the point of discharge. As site-specific water quality data is not available, in accordance with Table B implementing procedures,  $C_s$  equals zero for all pollutants, except the following:

**Table F-8. Background Concentrations—Ocean Plan**

| Pollutant | Background Seawater Concentration (ug/L) |
|-----------|------------------------------------------|
| Arsenic   | 3                                        |
| Copper    | 2                                        |
| Mercury   | 0.0005                                   |
| Silver    | 0.16                                     |
| Zinc      | 8                                        |

Applicable water quality objectives from Table B of the Ocean Plan are as follows.

**Table F-9. Water Quality Objectives—Ocean Plan**

| Pollutant               | Units | 6-Month Median | Daily Maximum | Instantaneous Maximum | 30 Day Average |
|-------------------------|-------|----------------|---------------|-----------------------|----------------|
| Total Residual Chlorine | µg/L  | 2              | 8             | 60                    | ---            |
| TCDD Equivalents        | µg/L  | ---            | ---           | ---                   | 3.9E-09        |
| Chloroform              | µg/L  | ---            | ---           | ---                   | 130            |
| Dichlorobromomethane    | µg/L  | ---            | ---           | ---                   | 6.2            |

Using the equation,  $C_e = C_o + D_m (C_o - C_s)$ , effluent limitations are calculated as follows. Here,  $D_m$  is equal to 100 for each effluent limitation calculation.

Total Residual Chlorine

$$C_e = 2 + 100 (2 - 0) = 202 \text{ µg/L (6-Month Median)}$$

$$C_e = 8 + 100 (8 - 0) = 808 \text{ µg/L (Daily Maximum)}$$

$$C_e = 60 + 100 (60 - 0) = 6060 \text{ µg/L (Instantaneous Maximum)}$$

TCDD Equivalents

$$C_e = 3.9\text{E-}09 + 100 (3.9\text{E-}09 - 0) = 3.94\text{E-}07 \text{ µg/L (30-Day Average)}$$

Chloroform

$$C_e = 130 + 100 (130 - 0) = 13130 \text{ µg/L (30-Day Average)}$$

Dichlorobromomethane

$$C_e = 6.2 + 100 (6.2 - 0) = 626 \text{ µg/L (30-Day Average)}$$

Mass-based effluent limitations have also been established for these pollutants pursuant to 40 CFR 122.45(f), which requires that pollutants limited in permits be

expressed in terms of mass. Mass-based limitations were established using the facility average dry weather design flow of 0.3 mgd. Review of facility discharge data from 1979 – 2007 show that average discharge rates have not exceeded the design capacity of 0.3 mgd in any season, and therefore the Regional Water Board has determined that year-round compliance with mass based limitations based on this design flow is achievable.

WQBELs established by the Order are summarized in the following tables and text.

**Table F-10. Final WQBELs for Ocean Plan Table B Pollutants**

| Pollutant               | Units                | 6-Month Median | Daily Maximum | Instantaneous Maximum | 30 Day Average |
|-------------------------|----------------------|----------------|---------------|-----------------------|----------------|
| Total Residual Chlorine | µg/L                 | 202            | 808           | 6060                  | ---            |
|                         | lbs/day <sup>5</sup> | 0.505          | 2.02          | 15.2                  | ---            |
| TCDD Equivalents        | µg/L                 | ---            | ---           | ---                   | 3.94E-07       |
|                         | lbs/day <sup>6</sup> | ---            | ---           | ---                   | 9.86E-10       |
| Chloroform              | µg/L                 | ---            | ---           | ---                   | 13,130         |
|                         | lbs/day <sup>5</sup> | ---            | ---           | ---                   | 32.9           |
| Dichlorobromomethane    | µg/L                 | ---            | ---           | ---                   | 626            |
|                         | lbs/day <sup>5</sup> | ---            | ---           | ---                   | 1.57           |

WQBELs for total coliform bacteria reflect the most stringent applicable water quality objectives of the Ocean Plan for protection of shellfish harvesting and are expressed as follows.

Disinfected effluent discharged through Discharge Point 001 shall not contain concentrations of total coliform bacteria, as measured at Monitoring Location EFF-002, exceeding the following limitations:

- (1) The median concentration shall not exceed a Most Probable Number (MPN) of 70 per 100 mL for a calendar month.
- (2) Not more than 10 percent of samples shall exceed an MPN of 230 per 100 mL, using bacteriological results from ten consecutive samples.

## 5. Whole Effluent Toxicity (WET)

Effluent limitations for whole effluent toxicity (WET), acute or chronic, protect the receiving water quality from the aggregate toxic effects of a mixture of pollutants in the effluent. There are two types of WET test – acute and chronic. An acute toxicity

<sup>6</sup> Mass-based limitations are based on the average dry weather design flow of the WWTF of 0.3 mgd.

test is conducted over a short time period and measures mortality. A chronic toxicity test is conducted over a longer period of time and may measure mortality, reproduction and/or growth. The Ocean Plan contains numeric water quality objectives for acute and chronic toxicity established in Table B.

The Discharger conducted chronic toxicity testing annually during the term of the previous permit, except during the first year of the permit term when testing was conducted at least semiannually, and utilized the red abalone (*Haliotis rufescens*), the topsmelt (*Atherinops affinis*), and the giant kelp (*Macrocystis pyrifera*). The Discharger was required to conduct accelerated testing if toxicity was detected above 7.6 chronic toxicity units (TUc). Maximum chronic testing results collected during the term of the previous permit are summarized in the table below. The Discharger has conducted accelerated monitoring each year since 2006.

**Table F-11. Chronic Toxicity Testing Results**

| Date           | Units | Results |
|----------------|-------|---------|
| January 2005   | TUc   | 6.25    |
| April 2005     | TUc   | 6.25    |
| September 2005 | TUc   | 6.25    |
| August 2006    | TUc   | 6.25    |
| September 2006 | TUc   | 12.50   |
| October 2006   | TUc   | 100.00  |
| November 2006  | TUc   | 12.50   |
| December 2006  | TUc   | 50.00   |
| January 2007   | TUc   | 100.00  |
| March 2007     | TUc   | 6.25    |
| April 2007     | TUc   | 6.25    |
| October 2007   | TUc   | 100.00  |
| November 2007  | TUc   | 6.25    |
| December 2007  | TUc   | 6.25    |
| May 2008       | TUc   | 100     |
| June 2008      | TUc   | 100     |
| July 2008      | TUc   | 6.25    |
| August 2008    | TUc   | 12.50   |

The facility indicated that the cause of elevated chronic toxicity results in 2008 was elevated residual chlorine concentrations, and has since installed a chemical feeder to dechlorinate the effluent with calcium thiosulfate prior to discharge. No reasonable potential was therefore found for chronic toxicity; however, continued annual monitoring for chronic toxicity is required by the MRP.

The trigger for accelerated monitoring in the previous permit was 7.6 TUc, which did not reflect the dilution factor of 100:1 that the facility outfall achieves. The Order establishes a new chronic toxicity trigger of 101 TUc, which takes into account this dilution ratio.

#### **D. Final Effluent Limitations**

##### **1. Satisfaction of Anti-Backsliding Requirements**

All effluent limitations in this Order are at least as stringent as the effluent limitations in the previous Order, except for copper, whose effluent limitations are not retained from the previous permit. Effluent limitations for copper were omitted from this Order based on new information, as discussed below. The Ocean Plan contains a procedure for determining reasonable potential and establishing effluent limitations. Eliminating effluent limitations from the previous permit that do not demonstrate reasonable potential meets the exception to antibacksliding requirements established at CWA 402(o)(2)(B)(ii). This exception states that a less stringent effluent limitation (here, the elimination of a limitation) may be included in a reissued permit when information is available which was not available at the time of permit issuance. New information considered by Regional Water Board staff to make a determination of “no reasonable potential” is effluent monitoring data generated since the previous permit was adopted.

##### **2. Satisfaction of Antidegradation Policy**

This Order is consistent with applicable federal and State antidegradation policies, as it does not authorize the discharge of increased concentrations of pollutants or increased volumes of treated wastewater.

##### **3. Stringency of Requirements for Individual Pollutants**

This Order contains both technology-based and water quality-based effluent limitations for individual pollutants. The technology-based effluent limitations consist of restrictions on BOD<sub>5</sub>, TSS, pH, settleable solids, oil and grease and turbidity at Discharge Point 001. Restrictions on these pollutants are discussed in section IV.B in this Fact Sheet. This Order’s technology-based pollutant restrictions are not more stringent than the minimum, applicable federal technology-based requirements. The Order also contains effluent limitations in addition to the minimum, federal technology-based requirements necessary to meet applicable water quality standards. These limitations are not more stringent than required by the CWA.

WQBELs have been scientifically derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. The scientific procedures for calculating the individual WQBELs

are based on the Ocean Plan, which was approved by USEPA on February 14, 2006. All beneficial uses and water quality objectives contained in the Basin Plan were approved under State law and submitted to and approved by USEPA prior to May 30, 2000. Any water quality objectives and beneficial uses submitted to USEPA prior to May 30, 2000 but not approved by USEPA by that date are nonetheless “applicable water quality standards for the purposes of the CWA pursuant to section 131.21(c)(1).” Collectively, this Order’s restrictions on individual pollutants are no more stringent than required by to implement the requirements of the CWA.

In addition, the Regional Water Board has considered all information submitted by the Discharger in considering the factors in Water Code section 13263, including the provisions of Water Code section 13241, in establishing these requirements.

#### **E. Interim Effluent Limitations**

This section of the standardized permit is not applicable to the Mendocino City CSD WWTF.

#### **F. Land Discharge Specifications**

This section of the standardized permit is not applicable to the Mendocino City CSD WWTF.

#### **G. Reclamation Specifications**

Since 1997, water reclamation at the Mendocino High School athletic fields has been regulated exclusively under Water Reclamation Requirements Order No. 97-66, issued jointly to the Discharger and the Mendocino Unified School District on August 27, 1997. In consultation with the California Department of Health Services (now California Department of Public Health), the Regional Water Board established in WDR Order No. 97-66 prohibitions, limitations, and requirements to comply with regulations for water reclamation contained in title 22, Division 4, Chapter 3 of the California Code of Regulations. Compliance with effluent limitations in WDR Order No. 97-66 for BOD<sub>5</sub>, suspended solids, total coliform organisms and turbidity is achieved through analyses of representative samples of effluent produced by the Discharger’s WWTF for use as reclaimed water. The effluent limitation requiring a minimum CT<sup>7</sup> value is met by calculation, using the product of chlorine residual and the minimum contact time within the storage tank, prior to irrigating the athletic fields at the Mendocino High School.

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<sup>7</sup> The CT value is the product of total chlorine residual and modal contact time measured at the same period. The modal contact time is the amount of time that elapsed between the time that a tracer, such as salt or dye, is injected into the influent at the entrance of the chlorination chamber and the time that the highest concentration of the tracer is observed in the effluent from the chamber.

In this renewed NPDES permit, effluent limitations for water reclamation where the Discharger has sole responsibility for compliance (i.e., BOD<sub>5</sub>, suspended solids, pH, total coliform organisms, turbidity, and filter overflow rate) are newly established in section IV.C of the Order. Use area requirements for water reclamation in title 22, Article 4, section 60310, remain in effect under provisions in WDR Order No. 97-66. Compliance with water reclamation limitations must be met at Monitoring Location EFF-001 for BOD, and suspended solids, at Monitoring Location EFF-002 for pH and total coliform organisms. The tertiary filter overflow rate is calculated using the WWTP flow as measured at INF-001 and known dimensions of the tertiary filters. Compliance with minimum turbidity requirement may be met using continuous turbidity monitoring at EFF-001. Compliance with these limitations will ensure that recycled water is adequately oxidized and filtered prior to discharge to the recycled water storage tanks. Reclamation specifications for minimum CT value, minimum contact time, and nitrate are also newly established in section IV.C of the Order. Compliance with these specifications may be achieved using measurements obtained at Monitoring Location REC-001 or other locations where samples can be collected that are representative of the treated wastewater to be reclaimed at the Mendocino High School athletic field. Where there is duplication of effluent limitations between WDR Order No. 97-66 and the NPDES permit, the Discharger shall comply with the more stringent limitation.

The new reclamation specification for nitrate found in section IV.C of the Order is included to comply with regulations contained in title 22, Division 4, Chapter 15 of the California Code of Regulations for the protection of public water supplies.

**Table F-12. Reclamation Specifications for Discharge Point 002**

| Parameter                         | Units               | Effluent Limitation |               |
|-----------------------------------|---------------------|---------------------|---------------|
|                                   |                     | Average Monthly     | Maximum Daily |
| BOD <sub>5</sub>                  | mg/L                | 10                  | 20            |
| TSS                               | mg/L                | 10                  | 20            |
| Nitrate (as N)                    | mg/L                | 10                  | ---           |
| pH                                | pH Units            | 6.0 – 9.0           |               |
| Total Coliform Organisms          | MPN/ 100 mL         | a                   |               |
| Turbidity                         | NTU                 | b                   |               |
| Filter Overflow Rate              | gpm/ft <sup>2</sup> | c                   |               |
| Disinfection Process Requirements | NTU                 | d                   |               |

- a. **Bacteria:** Disinfected effluent discharged to the water recycling system shall not contain coliform bacteria in excess of the following concentrations, with compliance measured at Monitoring Location EFF-002, as described in the attached MRP:

- (1) The median concentration shall not exceed a Most Probable Number (MPN) of 2.2 per 100 milliliters, using the bacteriological results of the last seven days for which analyses have been completed,
  - (2) The number of coliform bacteria shall not exceed an MPN of 23 per 100 milliliters in more than one sample in any 30-day period, and
  - (3) No single sample shall exceed an MPN of 240 total coliform bacteria per 100 milliliters.
- b. **Turbidity.** The effluent from the filtration system shall at all times be filtered such that the filtered effluent does not exceed any of the following specifications prior to discharge to the disinfection unit, with compliance measured at Monitoring Location EFF-001, as described in the attached MRP:
- (1) An average of 2 Nephelometric Turbidity Units (NTU) during a 24-hour period;
  - (2) 5 NTU more than 5 percent of the time during a 24-hour period; and
  - (3) 10 NTU at any time.
- c. **Filtration Rate.** The rate of filtration through the effluent filter shall not exceed five (5) gallons per minute per square foot of surface area or other filtration rates authorized in writing by the Executive Officer and under conditions recommended by the California Department of Public Health, with compliance determined using the WWTP flow measured at INF-001 and known dimensions of the effluent filters, as described in the attached MRP.
- d. **Disinfection Process Requirements.** Treated effluent shall be disinfected in a manner that ensures effective pathogen reduction as described in the following specifications, with compliance determined using the total chlorine residual measured at Monitoring Location REC-001, as described in the attached MRP:
- (1) Before recycled water is used to irrigate the athletic fields, a CT value<sup>8</sup> of not less than 450 milligram-minutes per liter shall be achieved by maintaining a chlorine residual in the recycled water storage tank for a predetermined contact time. The minimum allowable contact time, between the time pumping from the WWTF is stopped and the time irrigation begins, is 90 minutes
  - (2) Recycled water not meeting the CT criteria in section d(1), above, shall not be discharged for irrigation at the athletic fields.

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<sup>8</sup> The CT value is the product of total chlorine residual and modal contact time measured at the same period. The modal contact time is the amount of time that elapsed between the time that a tracer, such as salt or dye, is injected into the influent at the entrance of the chlorination chamber and the time that the highest concentration of the tracer is observed in the effluent from the chamber.

## **V. RATIONALE FOR RECEIVING WATER LIMITATIONS**

### **A. Surface Water**

CWA section 303(a-c) requires states to adopt water quality standards, including criteria where they are necessary to protect beneficial uses. The State Water Board adopted water quality criteria as water quality objectives in the Ocean Plan. The Ocean Plan includes numeric and narrative water quality objectives for various beneficial uses. This Order contains receiving surface water limitations based on the Ocean Plan numerical and narrative water quality objectives for dissolved oxygen, floating particulates, oil and grease, pH, discoloration, natural lighting, deposition of solids, dissolved sulfides, organic materials, and nutrient materials.

### **B. Groundwater**

Groundwater limitations included in the Order are necessary to protect existing groundwater quality and to protect the beneficial uses of groundwater.

1. The beneficial uses of the underlying ground water are municipal and domestic supply, industrial service supply, industrial process supply, agricultural supply, and freshwater replenishment to surface waters.
2. State Water Board Resolution No. 68-16, requires, in part, that whenever the existing quality of water is better than the quality established in policies as of the date on which such policies become effective, such existing high quality water will be maintained until it is demonstrated to the state that any changes will be consistent with maximum benefit to the people of the state, will not unreasonably affect beneficial uses of such water, and will not result in water quality less than prescribed in the policies.

## **VI. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS**

Section 122.48 requires that all NPDES permits specify requirements for recording and reporting monitoring results. Water Code sections 13267 and 13383 authorizes the Regional Water Board to require technical and monitoring reports. The Monitoring and Reporting Program (MRP), Attachment E of this Order, establishes monitoring and reporting requirements to implement federal and State requirements. The following provides the rationale for the monitoring and reporting requirements contained in the MRP for this facility.

### **A. Influent Monitoring**

Influent monitoring requirements for flow, BOD<sub>5</sub>, and TSS are retained from the previous permit and are necessary to determine compliance with the Order's 85 percent removal requirement for these parameters.

## **B. Effluent Monitoring**

Most effluent monitoring requirements for Discharge Point 001 at EFF-001 are retained from the previous permit. Changes in the MRP (Attachment E) from that of the previous permit are as follows:

- Routine monitoring for chloroform and dichlorobromomethane is established by the MRP to determine compliance with effluent limitations for these pollutants.
- The frequency of monitoring for oil and grease has been reduced from monthly to quarterly because monitoring data shows that most monitoring results have resulted in a non-detect concentration. Monitoring frequency for ammonia has been reduced from weekly to monthly because monitoring data shows that effluent concentrations are below the applicable water quality objective.
- The monitoring frequency for the Table B metals (antimony, arsenic, beryllium, cadmium, hexavalent and trivalent chromium, lead, nickel, selenium, silver, thallium, and zinc) and cyanide has been reduced from annually to once per five years, to be consistent with the monitoring requirements of the Ocean Plan, which state that for discharges less than 1 mgd, the monitoring frequency for Table B pollutants shall be once during the term of the permit.

Monitoring requirements for Discharge Point 002 (Water Reclamation) at EFF-001, EFF-002, and REC-001, are newly established in this Order. Changes in the MRP (Attachment E) from that of the previous permit are as follows:

- When discharging recycled water to the recycled water storage tank, the Discharger must monitor its treated effluent for BOD, suspended solids, total coliform organisms, pH, and total residual chlorine to demonstrate compliance with reclamation specifications in section IV.C of the Order. BOD and suspended solids are measured at the effluent wet well (EFF-001). Total coliform organisms and pH are measured in the chlorine contact tank (EFF-002).
- Before discharging recycled water from the recycled water storage tank to the athletic fields at Mendocino High School, the Discharger must determine the concentration of total chlorine residual and contact time in the recycled water storage tank to calculate the minimum CT value of recycled water and demonstrate compliance with reclamation specifications in section IV.C of the Order. Total residual chlorine is measured continuously through a flow meter installed in a 2-inch diameter pipe that conveys treated wastewater to the recycled water storage tank. Monitoring for nitrate is required to demonstrate compliance with reclamation specifications in section IV.C of the Order.

### **C. Whole Effluent Toxicity Testing Requirements**

Whole effluent toxicity (WET) limitations and monitoring requirements protect the receiving water quality from the aggregate effect of a mixture of pollutants in the effluent. Where the minimum initial dilution of the effluent ranges from 100:1 to 350:1, the Ocean Plan requires only chronic toxicity testing for the waste discharge. Because the outfall is designed to achieve a minimal dilution of 100:1, the Order includes only chronic toxicity testing. Quarterly monitoring for chronic toxicity is required to determine compliance with the Ocean Plan water quality objective for chronic toxicity. Chronic testing requirements are located in section V of the MRP. The dilution series required for chronic toxicity testing is different from that required by the previous permit and has been established to reflect the dilution provided by the outfall (100:1).

### **D. Receiving Water Monitoring**

#### **1. Surface Water**

Compliance with the receiving water limitation established in section V.A of the Order will be determined based on a physical, chemical, and biological survey of the outfall location; the survey is required once every 5 years. The requirement to implement the plan for a comparative evaluation of indigenous biota and submit a summary report at least 6 months prior to permit expiration is also retained from the previous permit.

#### **2. Groundwater**

The MRP does not establish groundwater monitoring requirements.

### **E. Other Monitoring Requirements**

This Order includes filtration process requirements for turbidity to determine compliance with requirements for recycled wastewater systems, established at California Code of Regulations title 22, division 4, chapter 3. In addition, filter surface loading rate requirements have been included in this Order to demonstrate compliance with recommendations in the California Department of Public Health 2007 Treatment Technology Report for Recycled Water.

The requirement to conduct an ocean outfall inspection once during the term of the permit is retained from the previous permit.

## **VII. RATIONALE FOR PROVISIONS**

### **A. Standard Provisions**

- 1. Federal Standard Provisions.** Standard Provisions, which apply to all NPDES permits in accordance with section 122.41, and additional conditions applicable to

specified categories of permits in accordance with section 122.42, are provided in Attachment D. The Discharger must comply with all standard provisions and with those additional conditions that are applicable under section 122.42.

Section 122.41(a)(1) and (b) through (n) establish conditions that apply to all State-issued NPDES permits. These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation to the regulations must be included in the Order. Section 123.25(a)(12) allows the state to omit or modify conditions to impose more stringent requirements. In accordance with section 123.25, this Order omits federal conditions that address enforcement authority specified in sections 122.41(j)(5) and (k)(2) because the enforcement authority under the Water Code is more stringent. In lieu of these conditions, this Order incorporates by reference Water Code section 13387(e).

- 2. Regional Water Board Standard Provisions.** In addition to the federal Standard Provisions (Attachment D), the Discharger shall comply with the Regional Water Board Standard Provisions provided in Standard Provisions VI.A.2.
  - a. Order Provision VI.A.2.a identifies the State's enforcement authority under the Water Code, which is more stringent than the enforcement authority specified in the federal regulations [e.g. 40 CFR sections 122.41(j)(5) and (k)(2)].
  - b. Order Provision VI.A.2.b requires the Discharger to notify Regional Water Board staff, orally and in writing, in the event that the Discharger does not comply or will be unable to comply with any Order requirement. This provision requires the Discharger to make direct contact with a Regional Water Board staff person.
  - c. Order Provision VI.A.2.c requires the Discharger to provide written certification that it has notified the State Office of Emergency Services and the local health officer or directors of environmental health within 24 hours after becoming aware of an unauthorized discharge to a drainage channel or a surface water. The Discharge is also required to provide written documentation of the circumstances of the spill event within five (5) days, unless the Regional Water Board waives the confirmation.

## **B. Special Provisions**

### **1. Reopener Provisions**

- a. Standard Revisions (Special Provisions VI.C.1.a).** Conditions that necessitate a major modification of a permit are described in section 122.62, which include the following:
  - (1) When standards or regulations on which the permit was based have been changed by promulgation of amended standards or regulations or by judicial

decision. If revisions of applicable water quality standards are therefore promulgated or approved pursuant to section 303 of the CWA or amendments thereto, the Regional Water Board will revise and modify this Order in accordance with such revised standards.

(2) When new information that was not available at the time of permit issuance would have justified different permit conditions at the time of issuance.

- b. Reasonable Potential (Special Provisions VI.C.1.b).** This provision allows the Regional Water Board to modify, or revoke and reissue, this Order if present or future investigations demonstrate that the Discharger governed by this Permit is causing or contributing to excursions above any applicable priority pollutant criterion or objective, or adversely impacting water quality and/or the beneficial uses of receiving waters.
- c. Whole Effluent Toxicity (Special Provisions VI.C.1.c).** This Order requires the Discharger to investigate the causes of, and identify corrective actions to reduce or eliminate effluent toxicity through a TRE. This Order may be reopened to include a limitation for a specific toxicant identified in the TRE.

## **2. Special Studies and Additional Monitoring Requirements**

### **a. Toxicity Reduction Evaluations (Provision VI.C.2.a).**

In addition to routine monitoring at Discharge Point 001 for chronic toxicity, this provision requires the Discharger to maintain and update, when necessary, a TRE Work Plan to ensure the Discharger has a plan to immediately move forward with the initial tiers of a TRE, in the event effluent toxicity is encountered in the future. The TRE is initiated by evidence of a pattern of toxicity demonstrated through the additional effluent monitoring provided as a result of an accelerated monitoring program.

**TRE Guidance.** The Discharger is required to prepare a TRE Work Plan in accordance with USEPA guidance. Numerous guidance documents are available, as identified below:

1. *Toxicity Reduction Evaluation Guidance for Municipal Wastewater Treatment Plants*, (EPA/833B-99/002), August 1999.
2. *Generalized Methodology for Conducting Industrial TREs*, (EPA/600/2-88/070), April 1989.
3. *Methods for Aquatic Toxicity Identification Evaluations: Phase I Toxicity Characterization Procedures*, Second Edition, EPA 600/6-91/005F, February 1991.

4. *Toxicity Identification Evaluation: Characterization of Chronically Toxic Effluents, Phase I*, EPA 600/6-91/005F, May 1992.
5. *Methods for Aquatic Toxicity Identification Evaluations: Phase II Toxicity Identification Procedures for Samples Exhibiting acute and Chronic Toxicity*, Second Edition, EPA 600/R-92/080, September 1993.
6. *Methods for Aquatic Toxicity Identification Evaluations: Phase III Toxicity Confirmation Procedures for Samples Exhibiting Acute and Chronic Toxicity*, Second Edition, EPA 600/R-92/081, September 1993.
7. *Methods for Measuring the Acute Toxicity of Effluents and Receiving Waters to Freshwater and Marine Organisms*, Fifth Edition, EPA-821-R-02-012, October 2002.
8. *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms*, Fourth Edition, EPA-821-R-02-013, October 2002.
9. *Technical Support Document for Water Quality-based Toxics Control*, EPA/505/2-90-001, March 1991

### **3. Best Management Practices and Pollution Prevention**

- a. **Pollutant Minimization Plan.** Provision VI.C.3.a is included in this Order pursuant to section III.C.9 of the Ocean Plan. A Pollutant Minimization Program is required when there is evidence that a toxic pollutant is present in effluent at a concentration greater than an applicable effluent limitation.

### **4. Construction, Operation, and Maintenance Specifications**

Section 122.41(e) requires proper operation and maintenance of permitted wastewater systems and related facilities to achieve compliance with permit conditions. An up-to-date operation and maintenance manual, as required by Provision VI.C.4.b of the Order, is an integral part of a well-operated and maintained facility.

### **5. Special Provisions for Municipal Facilities (POTWs Only)**

The Regional Water Board includes special provisions in all NPDES Orders for municipal wastewater treatment facilities regarding wastewater collection systems, sanitary sewer overflows, source control, sludge handling and disposal, operator certification, and adequate capacity. These provisions assure efficient and satisfactory operation of municipal wastewater collection and treatment systems.

**a. Wastewater Collection Systems (Provision VI.C.5.a)**

1. **Statewide General WDRs for Sanitary Sewer Systems.** The State Water Board issued General Waste Discharge Requirements for Sanitary Sewer Systems, Water Quality Order No. 2006-0003-DWQ (General Order) on May 2, 2006. The General Order requires public agencies that own or operate sanitary sewer systems with greater than one mile of pipes or sewer lines to enroll for coverage under the General Order. The General Order requires agencies to develop sanitary sewer management plans (SSMPs) and report all sanitary sewer overflows (SSOs), among other requirements and prohibitions.

Furthermore, the General Order contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows. Inasmuch as that the Discharger's collection system is part of the system that is subject to this Order, certain standard provisions are applicable as specified in Provisions, section VI.C.5 of the Order. The Discharger must comply with both the General Order and this Order. The Discharger and public agencies that are discharging wastewater into the facility were required to obtain enrollment for regulation under the General Order by December 1, 2006.

All NPDES permits for POTWs currently include federally required standard conditions to mitigate discharges [40 CFR 122.41(d)], to report non-compliance [40 CFR 122.41(1), (6), and (7)], and to properly operate and maintain facilities [40 CFR 122.41(e)]. This provision is consistent with these federal requirements.

2. **Sanitary Sewer Overflows and Sewage Spills.** The Order includes provisions (Provision VI.C.5.(a)(2), and Attachment D subsections I.C., I.D., V.E., and V.H.) to ensure adequate and timely notifications are made to the Regional Water Board and appropriate local, state, and federal authorities in case of sewage spills. In addition, as an Enrollee under General Order No. 2006-0003-DWQ, the Discharger is required to report SSOs to an online SSO database administered through the California Integrated Water Quality System (CIWQS) and via telefax when the online SSO database is not available. Detailed notification and reporting requirements for SSOs and sewage spills are specified in Attachment E subsection E (Monitoring and Reporting Program). The goal of these provisions is to ensure appropriate and timely response by the Discharger to SSOs to protect public health and water quality.

**b. Source Control (Provision VI.C.5.b)**

Because the average dry weather design flow of the facility is less than 5.0 mgd and because there are no significant industrial users discharging to the WWTF, the Order does not require the Discharger to develop a pretreatment program that conforms to federal regulations. The Order retains the requirement to perform certain source control functions to ensure that pollutants do not interfere with, pass through, or be incompatible with treatment operations, interfere with the use or disposal of sludge, or pose a health hazard to personnel.

**c. Sludge Disposal and Handling Requirements (Provision VI.C.5.c)**

The disposal or reuse of wastewater treatment screenings, sludges, or other solids removed from the liquid waste stream is regulated by sections 257, 258, 501, and 503, and the State Water Board promulgated provisions of title 27, California Code of Regulations.

The discharge of biosolids through land application is not regulated under this Order. Instead, the Discharger is required to obtain coverage under the State Water Board Order No. 2004-0012-DWQ, General Waste Discharge Requirements for the Discharge of Biosolids to Land as a Soil Amendment in Agricultural, Silvicultural, Horticultural, and Land Reclamation Activities (General Order) or other WDRs issued by the Regional Water Board prior to the application of biosolids to land.

Dried sludge generated at the Facility is currently hauled off-site to the Redwood Landfill in Novato.

**d. Operator Certification (Provisions VI.C.5.d)**

This provision, retained from the previous permit, requires the WWTF to be operated by supervisors and operators who are certified as required by title 23, California Code of Regulations, section 3680.

**e. Adequate Capacity (Provisions VI.C.5.e)**

This provision is retained from the previous permit. The goal of this provision is to ensure appropriate and timely planning by the Discharger to ensure adequate capacity for the protection of public health and water quality.

**6. Other Special Provisions**

- a. Storm Water. For the control of storm water discharged from the site of the wastewater treatment plant, the Discharge shall seek coverage under the State Water Board's Water Quality Order 97-03-DWQ, if applicable.

## **7. Compliance Schedules**

This section of the standardized permit is not applicable to the Mendocino City CSD WWTF.

## **VIII. PUBLIC PARTICIPATION**

The California Regional Water Quality Control Board, North Coast Region (Regional Water Board) is considering the issuance of waste discharge requirements (WDRs) that will serve as a National Pollutant Discharge Elimination System (NPDES) permit for the Mendocino City CSD WWTF. As a step in the WDR adoption process, the Regional Water Board staff has developed tentative WDRs. The Regional Water Board encourages public participation in the WDR adoption process.

### **A. Notification of Interested Parties**

The Regional Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Notification was provided through posting on the Regional Water Board's Internet site at: [http://www.waterboards.ca.gov/northcoast/public\\_notices/public\\_hearings/npdes\\_permits\\_and\\_wdrs.shtml](http://www.waterboards.ca.gov/northcoast/public_notices/public_hearings/npdes_permits_and_wdrs.shtml) and through publication in the Press Democrat on February 6, 2009.

### **B. Written Comments**

The staff determinations are tentative. Interested persons are invited to submit written comments concerning these tentative WDRs. Comments must be submitted either in person or by mail to the Executive Office at the Regional Water Board at the address above on the cover page of this Order.

To be fully responded to by staff and considered by the Regional Water Board, written comments must be received at the Regional Water Board offices by 5:00 p.m. on **March 5, 2009**.

### **C. Public Hearing**

The Regional Water Board will hold a public hearing on the tentative WDRs during its regular Board meeting on the following date and time and at the following location:

Date: April 29, 2010  
Time: 9 a.m., or as soon as possible thereafter as noticed in the final agenda  
Location: Ukiah Valley Conference Center  
200 South School Street  
Ukiah, CA

Interested persons are invited to attend. At the public hearing, the Regional Water Board will hear testimony, if any, pertinent to the discharge, WDRs, and permit. Oral testimony will be heard; however, for accuracy of the record, important testimony should be in writing.

Please be aware that dates and venues may change. Our Web address is <http://www.waterboards.ca.gov/northcoast> where you can access the current agenda for changes in dates and locations.

#### **D. Waste Discharge Requirements Petitions**

Any aggrieved person may petition the State Water Resources Control Board to review the decision of the Regional Water Board regarding the final WDRs. The petition must be submitted within 30 days of the Regional Water Board's action to the following address:

State Water Resources Control Board  
Office of Chief Counsel  
P.O. Box 100, 1001 I Street  
Sacramento, CA 95812-0100

#### **E. Information and Copying**

The Report of Waste Discharge (ROWD), related documents, tentative effluent limitations and special provisions, comments received, and other information are on file and may be inspected at the address above at any time between 8:30 a.m. and 4:45 p.m., Monday through Friday. Copying of documents may be arranged through the Regional Water Board by calling 707-576-2220.

#### **F. Register of Interested Persons**

Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the Regional Water Board, reference this facility, and provide a name, address, and phone number.

#### **G. Additional Information**

Requests for additional information or questions regarding this order should be directed to Charles Reed at 707-576-2752.